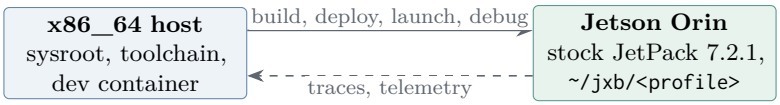
jxb

System Architecture and Internals

A reproducible cross-development, deployment and observability environment for ROS 2 on NVIDIA Jetson Orin



Technical documentation

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https://github.com/sansrlian/jetson-xbuild

Abstract

Developing ROS 2 software with CUDA, TensorRT and ONNX Runtime for embedded NVIDIA Jetson Orin modules is commonly done either directly on the board, which is slow and memory-bound, or in an emulated arm64 environment on a PC, which is slower still and breaks for CUDA. Cross-compilation avoids both, but it is only correct if the build sees exactly the libraries that the board will load at run time. This document describes jxb, an environment that (i) constructs the target root file system (*sysroot*) deterministically from the board's own package repositories, without emulation and without access to a board; (ii) compiles C++, CUDA and Python extension code with an x86_64 cross toolchain whose every component is pinned to the board's software release; (iii) deploys, runs, launches and debugs the result on unmodified, shared boards, either natively or in a GPU-enabled container derived from the same sysroot; and (iv) records and analyzes system behaviour with Nsight Systems, tegrastats, per-process sampling, kernel tracing and inference benchmarks, emitting machine-readable summaries. The design is inspired by the Renesas RZ/V2H RDK cross-build environment, whose structure it adopts; unlike the Jetson workflow previously derived from it, the sysroot is constructed from a declarative profile instead of being copied from a board. We describe the architecture and each mechanism in depth, report measurements on an Orin Nano 8 GB with JetPack 7.2.1 (for example, a 3.2× lower latency of the SAM detector model with a TensorRT fp16 engine than with ONNX Runtime's CUDA execution provider, and the full SAM autonomy stack running at 2.8 CPU cores, 2.1 GB and 8.4 W), and discuss the problems encountered during the implementation (62 recorded findings) and how they were resolved.

How to read this document. Sections 1 and 2 motivate the system and introduce the concepts it relies on; Section 3 gives the architecture overview; Sections 4 to 8 are deep dives into the four subsystems and the editor integration; Section 9 evaluates the system on the SAM stack; Sections 10 and 11 discuss challenges and the implementation process. For day-to-day use, see `docs/getting-started.md`; the engineering log `docs/findings.md` is cited as `Fn`.

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1 Introduction

1.1 Motivation

The system was built for the SAM project, an autonomous model vehicle whose ROS 2 stack (about 27 packages: camera drivers, image processing with CV-CUDA, a YOLO object detector on ONNX Runtime, localization, an acados-based trajectory planner and control) runs on NVIDIA Jetson Orin modules. Before `jxb`, the team used a setup derived from the Renesas RZ/V2H RDK [1], which had five recurring problems (Table 1).

Table 1: Problems of the previous workflow.

	Problem	Effect
P1	The sysroot was synchronized (`rsync`) from a live board	not reproducible; drifts with every board; needs a prepared board before anything can be built
P2	CUDA code was built in an emulated (QEMU) arm64 chroot	slow; `nvcc` and linking often failed, so CUDA packages were built on the board
P3	Distribution mismatch	the stack targets ROS 2 Lyrical (Ubuntu 26.04), the board runs JetPack 7.2.1 (Ubuntu 24.04)
P4	No multi-board view	no common deploy, trace collection or health view
P5	CUDA minor-version drift	the desktop development image has CUDA 13.3, the board 13.2; TensorRT engines are version-bound

1.2 Requirements

From these problems and from the operating conditions (one shared lab board, several developers, later a fleet), the following requirements were derived:

R1 *Reproducibility.* The build environment is a function of a declarative profile, not of a board's history.

R2 *Exact ABI match.* Every header and library used at build time is the version the board loads.

R3 *No emulation for compilation.* C++, CUDA and Python extensions are compiled by native x86_64 compilers.

R4 *Boards stay stock.* No `apt` changes, no writes outside `~/jxb/`, no root; boards are shared.

R5 *Several ROS 2 distributions* (Jazzy, Kilted, Lyrical) on the same stock board, switchable per command.

R6 *One workflow, three front ends:* terminal, VS Code and, later, a GUI run the same operations.

R7 *Observability with machine-readable results* (CPU, memory, GPU, power, scheduling, inference).

R8 *Safety on shared hardware.* Measurements must not overlap; leftovers must not keep running.

1.3 Contributions

1. A **sysroot builder** that resolves the arm64 package closure with a private `apt` instance against the board's repositories and reproduces the relevant side effects of maintainer scripts without executing them (Section 4).
2. A **CMake/colcon cross toolchain** for C, C++, CUDA and Python extensions, including the corrections that CUDA 13.2, CMake and ROS 2 need for host builds (Section 5).

3. A **board runtime model** that makes stock boards run other ROS distributions, either by shipping a relocatable ROS tree plus missing libraries (found by an `ldd` fix-point iteration) or by running a container image created from the sysroot itself (Section 6).
4. A **remote launch mode** in the style of the Renesas RDK, with process-group lifecycle management, monitoring and bag recording, and SSH-tunnelled remote debugging (Sections 6.6 and 6.7).
5. An **observability pipeline** that turns Nsight Systems databases, tegrastats logs, `/proc` samples and ftrace text into one `summary.json` per recording (Section 7).
6. An **evaluation** on the SAM stack and a catalogue of findings, several of them upstream bugs (Sections 9 and 10).

1.4 Conventions

Host is the x86_64 PC, *board* the Jetson. Paths such as `~/jxb/<profile>` are board paths; `sysroots/<profile>` is relative to the repository on the host. `<profile>` stands for a profile name such as `jp721-kilted`. Code excerpts are taken from the repository and abridged where marked with `...`.

2 Background and concepts

This section introduces the concepts the rest of the document relies on. Readers familiar with cross-compilation, JetPack and colcon may skip to Section 3.

2.1 Cross-compilation in a nutshell

In GNU terminology, the *build* machine runs the compiler and the *host* machine runs the produced program; embedded practice, and this document, call the PC "host" and the device "board" or "target". Producing correct binaries for another machine requires two ingredients:

1. A **toolchain** that emits code for the target instruction set and ABI: here `aarch64-linux-gnu-gcc` (GCC 13, the default compiler of Ubuntu 24.04) and NVIDIA's `nvcc`.
2. A **sysroot**: a directory that looks like the target's root file system for the purposes of compiling and linking. The compiler finds headers in `<sysroot>/usr/include`, the linker finds libraries in `<sysroot>/usr/lib/aarch64-linux-gnu`, and both treat `<sysroot>` as `/`.

Linking against a shared library records information that the board's dynamic loader (`ld-linux-aarch64.so.1`) later relies on [16]. These records explain most cross-build failures discussed in Section 10 (Table 2).

Table 2: ELF records that a cross build must get right.

Record	Meaning	Typical cross-build failure
`DT_NEEDED`	names of the libraries to load, normally their `SONAME` (e.g. `libcudart.so.13`)	an absolute host path is recorded when a library has no `SONAME` (F44)
`DT_RUNPATH`	search directories baked into the binary	host paths such as `/home/...` leak in
version needs	required symbol versions such as `GLIBC_2.38`, `GLIBCXX_3.4.32`	built against a newer glibc or libstdc++ than the board has
`PT_INTERP`	path of the dynamic loader	wrong loader for the target
ISA features	instruction-set extensions used by the code	`SIGILL` on a CPU without them (F25)

Definition

A sysroot is *correct* for a board if, for every file the build reads, the sysroot's version is identical to the version installed on the board. jxb's central claim is that correctness can be established *by construction* (Section 4) instead of by copying.

2.2 The Jetson Orin platform and JetPack 7.2.1

The Orin modules combine Arm Cortex-A78AE cores with an NVIDIA Ampere GPU of compute capability 8.7 (`sm_87`) that shares the LPDDR5 memory with the CPU. All measurements in this document were taken on an **Orin Nano 8 GB** (6 cores, 7.3 GB usable memory, 25 W power mode, NVMe storage) running a **PREEMPT_RT** kernel (`6.8.12-rt-tegra`).

JetPack 7.2.1 is NVIDIA's software release for it [6]: Jetson Linux (L4T) 39.2.1 on Ubuntu 24.04, CUDA 13.2 (`cudart` 13.2.86), cuDNN 9.20, TensorRT 10.16.2, VPI 4.1 and an NVIDIA build of OpenCV 4.8 (`F1`). It is installed with `apt` from three NVIDIA repositories (`common`, `som` and `ffmpeg` of release `r39.2` at `https://repo.download.nvidia.com/jetson/`), which an apt preference pins above Ubuntu's packages with priority 600. Two platform details become important later:

- CUDA 13 uses the unified Arm ("SBSA") layout `targets/sbsa-linux` on Orin, not `aarch64-linux` as in JetPack 6 (`F2`).
- The CUDA driver is installed per GPU variant below `/opt/nvidia/l4t-gpu-libs/` (`nvgpu` for Orin, `openrm` for Thor); a boot service publishes the active one (`F26`). An extracted sysroot therefore has no `libcuda.so.1` until that step is replayed.

2.3 ROS 2 builds: colcon, ament and code generators

A ROS 2 workspace is built by **colcon** [11], which builds each package (CMake or Python) in dependency order and installs it into an *install space*. Packages find each other through `CMAKE_PREFIX_PATH`, `AMENT_PREFIX_PATH` and the ament resource index; at run time, generated `setup.bash`/`local_setup.bash` scripts put the install space on `PATH`, `LD_LIBRARY_PATH` and `PYTHONPATH`. The ROS installation itself (`/opt/ros/<distro>`) is the *underlay* [12].

Cross-building ROS 2 is harder than cross-building a plain CMake project because **code generators run during the build**: `rosidl` turns `.msg` files into C, C++ and Python sources, ament writes CMake configuration files. These generators are Python programs that must run on the build machine, while the code they emit is compiled for the target. jxb therefore installs ROS 2 twice: on the host for the generators and in the sysroot for the target libraries, both from the same distribution sync.

2.4 The CUDA compilation model

`nvcc` [7] splits a `.cu` file into device code, which it compiles to GPU machine code (SASS) for an architecture such as `sm_87` and optionally to PTX for just-in-time compilation, and host code, which it passes to a host C++ compiler (`-ccbin`). Because `nvcc` itself is an x86_64 program that needs only the target's headers and libraries for the host part, it can cross-compile: with `-ccbin aarch64-linux-gnu-g++` and `--target-directory sbsa-linux`, the device code is identical to a build on the board and the host code is aarch64. This removes problem P2.

2.5 The reference: the Renesas RZ/V2H RDK cross-build environment

Renesas documents a cross-build setup for its RZ/V2H RDK that served as the model for jxb's structure [1, 2, 3, 4]. Its development guide introduces the concepts and limitations of cross-compiling, then walks through environment setup (a Docker image with an arm64 sysroot baked

into `/opt/arm64_sysroot`), a usage guide (workspace layout, an arm64 chroot via QEMU for `rosdep`, `cross-colcon-build`), a VS Code workspace with tasks for building, deployment and remote debugging via `gdbserver`, a tutorial, known issues (for example, non-relocatable CMake paths inside the sysroot) and an FAQ. Numbered procedures, notes and tips make it easy to follow. This document borrows that structure: concepts first, then the architecture, then each workflow, with known issues and an FAQ at the end.

jxb keeps the RDK's good ideas (one command per workflow, a long-lived build container, VS Code tasks that wrap CLI commands, gdbserver debugging) and changes what did not scale for a Jetson fleet. The SAM team's earlier Jetson workflow, derived from the RDK, synchronized its sysroot from live boards and built CUDA code in the emulated chroot (P1, P2). Table 3 compares the RDK image and that derived workflow with jxb.

Table 3: The RDK-based setup and jxb.

Aspect	RDK-based setup	jxb
Sysroot origin	prebuilt, baked into the image (14.5 GB image); for Jetson synchronized from boards	built from the board's apt repositories, per profile
Emulation	arm64 chroot (QEMU) for `rosdep`; in the Jetson workflow also for CUDA builds	none for compilation; QEMU only for optional smoke tests
CUDA	not part of the RDK (other SoC)	x86 `nvcc` of the board's exact version, `sm_87`
Distributions	one (Jazzy)	Jazzy, Kilted, Lyrical on 24.04, Lyrical on 26.04
Boards	one configured target, password-based SSH (`sshpass`)	board inventory, SSH keys only, nothing installed on boards
Run-time modes	native	native or container, per command
Observability	not part of the cross-build guide	Nsight Systems, tegrastats, `/proc` sampling, ftrace, benchmarks

3 System architecture

3.1 Overview

jxb consists of a host-side command-line interface and the artefacts it manages on the host and on the boards (Fig. 1). The host holds four artefacts per profile: the **xbuild image** (host tools only, ≈3.4 GB), the **sysroot** (target files only, 16–19 GB), the **dev container** that combines both with the developer's workspace, and, for container mode, the **runtime image** (the sysroot without build-only files). Boards receive only what a run needs, below `~/jxb/<profile>/`.

3.2 Components

Table 4 lists the components with their location and role. The CLI uses only the Python standard library, so that it runs on any developer PC; everything that needs more (matplotlib, ROS, compilers) runs in the dev container or on the board.

3.3 Profiles: the single source of truth

A **profile** is a TOML file that pins everything that must match between the xbuild image, the sysroot and the board: the JetPack/L4T version, CPU flags, CUDA version and architecture, the ROS distribution, the apt sources and pins, the package sets and the extras (Listing 1). Profiles inherit from one another; tables are merged recursively, all other values are replaced (Fig. 2):

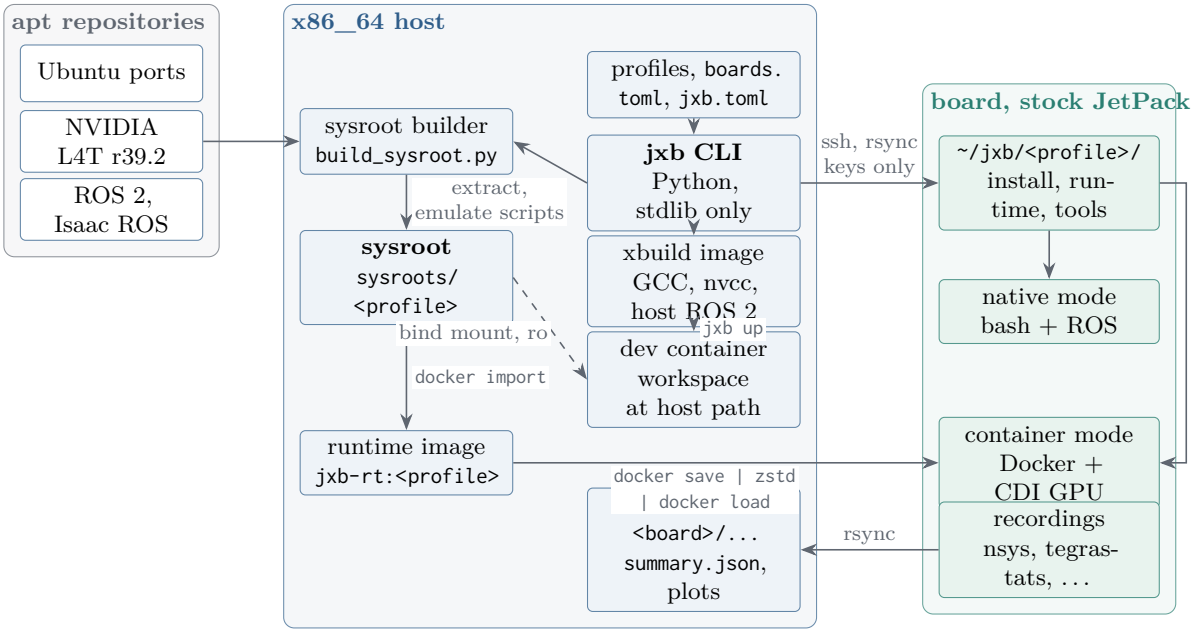


Figure 1: Components and data flows. Solid arrows produce or transfer data; the dashed arrow is a read-only bind mount. The sysroot builder and the dev container both run in the xbuild image.

```python
def _merge(base: dict, override: dict) -> dict:
    out = dict(base)
    for k, v in override.items():
        out[k] = _merge(out[k], v) if isinstance(v, dict) and isinstance(out.get(k), dict) else v
    return out
```

```toml
[board]
jetpack = "7.2.1"
l4t = "39.2.1"
cpu_flags = "-mcpu=cortex-a78ae"
cuda_arch = "87"
gpu_variant = "nvgpu"              # JetPack 7.2 CUDA driver variant; Thor: openrm
ros_distro = "jazzy"

[image]
cuda_version = "13.2"
cuda_pkg_version = "13.2.86-1"     # must equal the board's cuda-cudart-13-2

[sysroot]
codename = "noble"
cuda_target = "sbsa-linux"
sources = [
    "deb [signed-by={ubuntu_key}] http://ports.ubuntu.com/ubuntu-ports noble main restricted universe
    ↪ multiverse",
    "deb [signed-by={nvidia_key}] https://repo.download.nvidia.com/jetson/common r39.2 main",
    ...
    "deb [signed-by={ros_key}] http://packages.ros.org/ros2/ubuntu noble main",
]
package_sets = ["base-dev", "runtime-base", "jetpack-7.2.1-dev", "l4t-driver", "ros-dev"]
extras = ["onnxruntime", "cvcuda", "acados"]
```

Listing 1: Excerpt of the base profile `profiles/jp721-jazzy.toml`.

Table 4: Components of jxb.

Component	Location	Runs on	Role
CLI	`jxb`	host	images, sysroot, container, build, boards, deploy, run, debug
launch mode	`toolchain/jxb_launch.py`	host→board	deploy, `ros2 launch`, monitors, bags, helpers
tracing	`toolchain/jxb_trace.py`	host→board	nsys, tegrastats, ftrace, benchmarks, analysis
plots	`toolchain/jxb_plot.py`	container	plots, `report.md`, comparisons
VS Code	`toolchain/jxb_vscode.py`	host	tasks and debug configurations
profile loader	`toolchain/jxb_profile.py`	host, container	TOML with `inherit`
xbuild image	`docker/xbuild.Dockerfile`	Docker	cross GCC, nvcc, host ROS 2, generators
sysroot builder	`sysroot/build_sysroot.py`	image	private apt, extraction, script emulation
package sets	`sysroot/sets/`	–	what the sysroot contains
fixups	`sysroot/fixups.toml`	–	patches for files that only break on the host
CMake toolchain	`toolchain/orin.toolchain.cmake`	container	C/C++/CUDA/Python cross settings
environment	`toolchain/env.sh`, `profile_env.py`, `ros-cross-env.sh`	container	variables derived from the sysroot's profile
build wrapper	`toolchain/bin/cross-colcon-build`	container	colcon with the toolchain, post-processing
binary checks	`jxb-elfcheck`, `jxb-fix-needed`	container	linkage verification, `DT_NEEDED` repair
dependency scan	`toolchain/bin/jxb-rosdeps`	image	rosdeps and launch-file dependencies
extras	`extras/`	container	ONNX Runtime GPU, acados, depthai
board tools	`tools/board/`	board	ftrace helper, benchmark, synthetic camera, bag→CSV
tests	`tests/`	all	end-to-end check of a profile

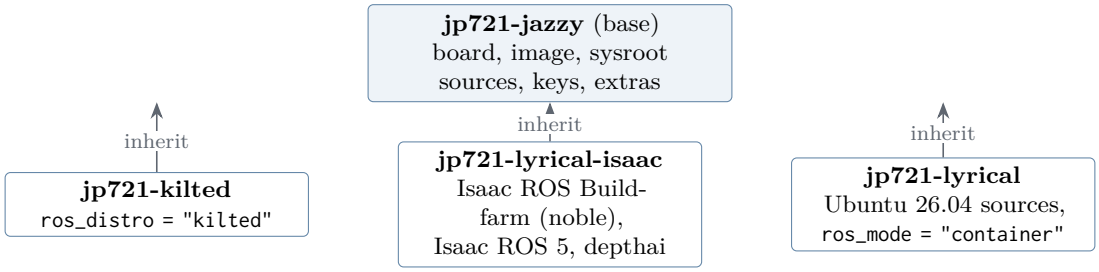


Figure 2: Profile inheritance: each child overrides only what differs from the base profile.

Table 5: Profiles and how they run on a stock JetPack 7.2.1 board (which has ROS 2 Jazzy).

Profile	ROS 2 / userland	On the board
`jp721-jazzy`	Jazzy / Ubuntu 24.04	the board's own ROS
`jp721-kilted`	Kilted / Ubuntu 24.04	shipped to `~/jxb/jp721-kilted/runtime`, or container
`jp721-lyrical-isaac`	Lyrical (NVIDIA Buildfarm) + Isaac ROS 5.0 / Ubuntu 24.04	shipped, or container
`jp721-lyrical`	Lyrical / Ubuntu 26.04	container only (glibc 2.43)

The resolved profile is written into the sysroot (`.jxb/profile.json`) when the sysroot is built,

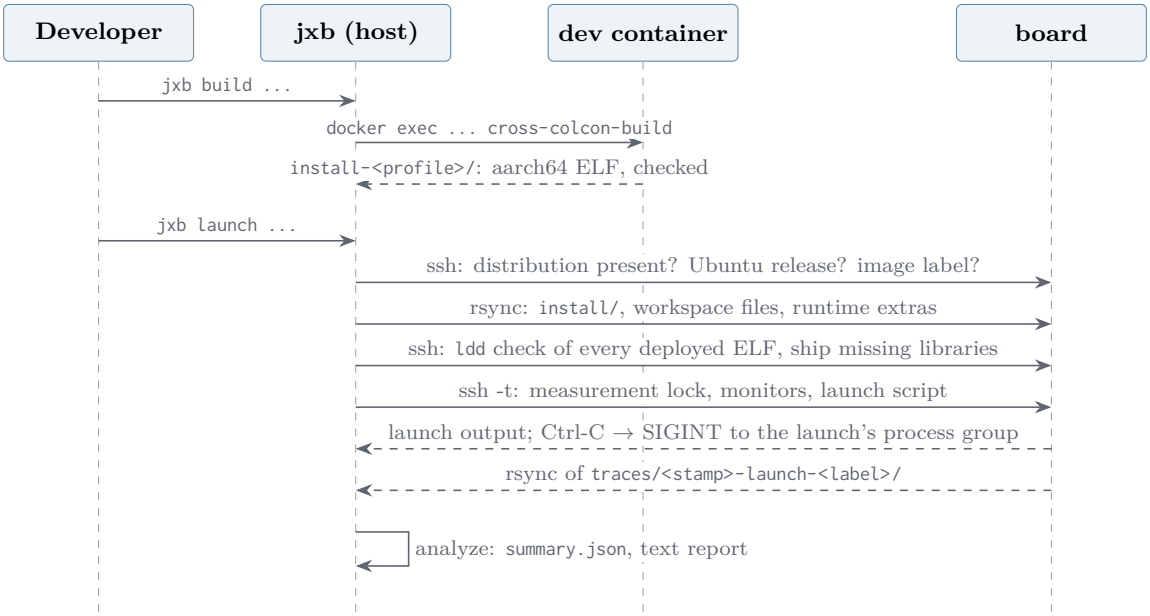


Figure 3: A build, launch and analysis session.

and every later step reads it from there. A sysroot and the settings it was built with can therefore never diverge (Section 5.2).

3.4 Design principles

D1 Same repositories, same versions. The sysroot is installed from the repositories, keys and pins the board itself uses, so its package versions are the board's by construction.

D2 Extract, do not execute. Packages are unpacked; the few maintainer-script effects that matter for compiling and linking are reproduced in Python. No QEMU, no root, no board.

D3 Host paths everywhere. Workspaces and the sysroot are mounted into the container at their host paths, so `compile_commands.json`, debug information and error messages are valid on the host and in the container alike.

D4 The board is borrowed, not owned. Everything `jxb` deploys lives below `~/jxb/`; what the board lacks is shipped next to the deployment, never installed system-wide.

D5 Every workflow is one non-interactive command with a JSON form of its result (`--json`, `summary.json`), so VS Code, scripts and a future GUI can drive the same code.

D6 Fail loudly and early. Missing keys, unresolved rosdeps, unbuilt packages, missing libraries and wrong ELF attributes stop the workflow with a message that says what to do.

3.5 Anatomy of a session

Figure 3 shows what happens between typing `jxb build` and reading a performance report; Sections 4 to 7 explain each step.

4 Deep dive I: the sysroot builder

4.1 Problem statement

The sysroot must contain the headers, libraries, CMake and pkg-config files and Python modules of every package the workspace builds against, **in the board's versions**, and in a form the host



Figure 4: The sysroot build pipeline. No step executes target code.

toolchain can use. Copying it from a board (P1) violates R1 and R4: the result depends on the board's history, and it requires a board. Installing the packages into an arm64 chroot with `apt` under QEMU is reproducible but slow, and it needs root, because `dpkg` runs maintainer scripts.

The observation behind the builder is that **compiling and linking need only the files of a package and very few side effects of its scripts**. Package *resolution* can be done by `apt` running natively on the host if `apt` is told that arm64 is its native architecture; *extraction* is a `tar` operation; and the relevant script effects can be enumerated (Section 4.5). Figure 4 gives the complete pipeline.

4.2 A private apt instance

`PrivateApt` creates a self-contained apt configuration in the profile's cache: its own `sources.list` (the profile's `sources` with the downloaded keyrings substituted for `{nvidia_key}` and the like), its own preferences (the board's pin), package lists, an archive cache and an **empty dpkg status file**. With `APT_CONFIG` pointing at the file in Listing 2, `apt-get` behaves as if it managed an empty arm64 system.

```
Dir "<cache>/apt/";
Dir::State::status "<cache>/apt/var/lib/dpkg/status";      // empty: nothing is installed
Dir::Cache::Archives "<deb-cache>/";                       // shared between profiles
APT::Architecture "arm64";
APT::Architectures { "arm64"; };
APT::Install-Recommends "false";
APT::Install-Suggests "false";
Debug::NoLocking "true";                                   // no root needed
```

Listing 2: Configuration of the private apt instance (generated).

Two details are easy to get wrong. First, apt's `signed-by` option decides the key format by file extension (`.asc` for ASCII-armored keys, `.gpg` for binary keyrings), while the ROS key is a binary keyring served under a neutral name; the builder therefore names downloaded keys by their content (F16). Second, the pin must be exactly the board's: NVIDIA's packages win with priority 600, except that the profiles pin Ubuntu's `libopencv-dev` at 700, because ROS `cv_bridge` links Ubuntu's OpenCV 4.6 and mixing it with NVIDIA's 4.8 made every `cv_bridge` node load two OpenCV versions (F17).

Why it matters

The preference block is copied from `nvidia-l4t-apt-source`, the package that configures apt on a JetPack board. This is what makes design principle D1 hold: apt applies identical inputs to identical candidate sets and therefore selects identical versions.

4.3 Resolving the closure

The requested packages are the union of the profile's package sets (`sysroot/sets/*.txt`, with `{ros_distro}` substituted) and the workspace set generated by `jxb-rosdeps` (Section 4.9). Resolution uses apt's own solver in simulation mode, and the selection is parsed from its output:

```
INST_RE = re.compile(r"^Inst (\S+) (?:\[\S+\] )?\((\S+) .*\[(\S+)\]\)")

def resolve(self, packages: list[str]) -> list[tuple[str, str, str]]:
    out = self.apt("-s", "install", *packages, capture=True)
    sel = [m.groups() for m in map(INST_RE.match, out.splitlines()) if m]   # (name, version, arch)
    ...
```

A line of the form `Inst libnvinfer10 (10.16.2.10-1+cuda13.2 <origin> [arm64])` yields name, version and architecture. Because the dpkg status is empty, the simulation returns the **complete transitive closure**: 944 packages for Jazzy, 995 for Kilted, 1068 for Lyrical with Isaac ROS and 1100 for Lyrical on Ubuntu 26.04. `apt-get install --download-only` then fetches exactly these files into the shared `.deb` cache; the first profile downloads about 8 GB.

Packages of a workspace set that no configured repository provides (for example `depthai` for Lyrical on Ubuntu 24.04) are removed with a warning before the resolution (`apt-cache policy` shows no candidate), so a single missing optional dependency does not fail the whole sysroot.

4.4 Extraction

A `.deb` file is an `ar` archive with a `control.tar` and a `data.tar`. The builder streams the data archive into the staging directory and keeps the control files for Section 4.5:

```
def extract_deb(deb: Path, dest: Path, scripts_dir: Path) -> str:
    data = subprocess.Popen(["dpkg-deb", "--fsys-tarfile", str(deb)], stdout=subprocess.PIPE)
    tar = subprocess.run(["tar", "-x", "-C", str(dest), "--keep-directory-symlink", "--no-same-owner",
                          "--warning=no-timestamp", "-f", "-"], stdin=data.stdout, ...)
    ...
    run(["dpkg-deb", "-e", str(deb), str(ctrl_dir)])        # control files: postinst, ...
    return (ctrl_dir / "control").read_text()
```

`--keep-directory-symlink` matters because Ubuntu uses a **merged `/usr`**: `/lib`, `/bin` and `/sbin` are symbolic links into `/usr`. The builder creates these links first, and `tar` must follow them instead of replacing them with directories; otherwise `/lib/aarch64-linux-gnu` and `/usr/lib/aarch64-linux-gnu` would become two different directories. Extraction runs in a thread pool with one job per CPU; a complete rebuild from the cache takes about one minute.

Algorithm 1 Replaying `update-alternatives` (dpkg's automatic mode).

best $\leftarrow \emptyset$ ▷ group name $\mapsto$ (link, path, priority, slaves)
for each `postinst` script s of the extracted packages **do**
  for each literal call `--install` $(\ell, g, p, \pi, \mathit{slaves})$ in s without shell variables **do**
    if p exists in the sysroot **and** $(g \notin \mathit{best}$ **or** $\pi > \mathit{best}[g].\pi)$ **then**
      $\mathit{best}[g] \leftarrow (\ell, p, \pi, \mathit{slaves})$
    end if
  end for
end for
for each $(g, (\ell, p, \pi, \mathit{slaves})) \in \mathit{best}$ **do**
  create relative links $\ell \rightarrow$ `/etc/alternatives/`$g \rightarrow p$
  do the same for every slave whose path exists
end for

4.5 Emulating maintainer scripts

`dpkg` runs `preinst` and `postinst` scripts that may create files the build needs. Executing them is exactly what `jxb` avoids (they are arm64 programs or call them, and often need root). Instead, the builder reproduces the four effects that matter for building and linking [15]:

1. **The dpkg status database.** Some build tools ask dpkg what is installed. The builder writes every control paragraph with `Status: install ok installed` into `var/lib/dpkg/status`.
2. **`update-alternatives` links.** Many `-dev` packages register alternatives (BLAS/LAPACK, `cpp`). The builder parses literal calls `update-alternatives --install LINK NAME PATH PRIO [--slave ...]` from all `postinst` scripts and creates the links of the winning candidate (Algorithm 1).
3. **Literal `ln -s` in `postinst`.** CV-CUDA and VPI create their CMake configuration links and `ld.so.conf.d` entries with plain `ln -s`, partly through shell variables (`F13`). The builder replays `ln -s SRC DEST` lines and expands variables that the script assigns literally (`VAR="..."`); a line with an unknown variable is skipped rather than guessed.
4. **The GPU driver variant.** At boot, JetPack's `nv-load-gpu-libs.sh` publishes the active CUDA driver variant (`F26`). The builder writes the same `/etc/ld.so.conf.d/000-nvidia-gpu-libs.conf` and the `libcuda.so` links for the profile's `gpu_variant` (Orin: `nvgpu`).

Finally, because CMake versions before 4 only look for `targets/aarch64-linux` when cross-compiling CUDA for aarch64, the builder adds that name as an alias of `targets/sbsa-linux` (`F2`).

Limitation

The selection reproduces dpkg's *automatic* mode. A board can still differ if someone selected an alternative manually: on the Nano, BLAS and LAPACK point to the reference implementations while the sysroot selects OpenBLAS (`F43`). Both have the same soname, so programs run, but gdb shows wrong symbols for these two libraries.

4.6 Relative symbolic links

Packages contain absolute symbolic links such as `libfoo.so` $\rightarrow$ `/usr/lib/aarch64-linux-gnu/libfoo.so.1`. Inside the sysroot on the host, such a link points into the *host's* `/usr` and silently links host libraries. The builder rewrites every absolute link target t of a link in directory d into

$$t' = \mathrm{relpath}(\mathrm{root} \oplus t,\ d), \tag{1}$$

where $\oplus$ joins the sysroot root and the board path. The links then resolve inside the sysroot on the host, and they still resolve correctly when the tree is used as a root file system (the runtime image,

Section 6.5). Links created for alternatives are computed against the *physical* parent directory, because `/lib/cpp` physically lives in `usr/lib` once `lib` → `usr/lib`.

4.7 Fixups

A few sysroot files are correct on the board but break when used from the host. Declarative rules in `sysroot/fixups.toml` patch them; each rule names the file, a search string, its replacement and the reason (Table 6).

Table 6: Fixups applied to sysroot files.

Rule	Why it is needed
`rosidl_adapt_interfaces.cmake`: `file(STRINGS ... ENCODING UTF-8)`	without the encoding, CMake splits lines at non-ASCII bytes, so any workspace below a path such as `.../HochschuleMünchen/...` fails to generate interfaces (F11, a ROS 2 bug)
`acados_template`: return after the build if `JXB_CROSS_BUILD` is set	acados generates, compiles and then `dlopen()`s the solver; an aarch64 library cannot be loaded by the x86 build process, and the failed load deleted the freshly built library (F20)
CUB `device_transform.cuh`, `tuning_transform.cuh`: specialization inside its namespace	a CUDA 13.2 bug [9]: GCC 13/14 reject the qualified class-head name that nvcc's front end re-emits; it breaks ONNX Runtime's CUDA EP, also natively (F22)

4.8 Extras

Some software is not available as a suitable apt package:

- **ONNX Runtime 1.31 with CUDA and TensorRT execution providers for `sm_87`.** The released aarch64 CUDA EP plugin contains code only for `sm_89/90a/120a/121a` and no PTX, so it cannot run on Orin (F8). jxb cross-builds ONNX Runtime (`extras/onnxruntime/build.sh`) in a "lean" flavour that keeps the contrib operators (the TensorRT EP requires them) but drops the LLM-specific kernel families; a full build was estimated at about 7 h on the 20-core workstation (F24). The sysroot uses the local build automatically when it exists, else the CPU release; the release's CMake configuration expects `lib64/` and `include/onnxruntime/`, which the builder adds as links (F9).
- **CV-CUDA 0.16** from its release packages (by URL), **acados 0.5.2** and **depthai 3.10** cross-built by jxb; depthai through vcpkg with an overlay triplet that chainloads the jxb toolchain (F34, F40).

4.9 Package sets and workspace dependencies

The sysroot's content is declared in package sets (Table 7).

Table 7: Package sets.

Set	Content
`base-dev`	`build-essential` (libc and libstdc++ headers, CRT files), Python headers and numpy, Eigen, yaml-cpp, Boost, OpenSSL, zlib, protobuf, ...
`runtime-base`	Ubuntu's *Essential* packages (`bash`, `coreutils`, `findutils`, `grep`, ...), which apt never pulls in as dependencies but a container needs (F48)
`jetpack-7.2.1-dev`	CUDA 13.2 libraries and the target headers `nvcc` needs (pinned), cuDNN, TensorRT, VPI, ...
`l4t-driver`	`nvidia-l4t-core`, `nvidia-l4t-cuda`, the `nvgpu` CUDA driver
`ros-dev`	`ros-<distro>-ros-base`, generators, Cyclone DDS, `cv_bridge`, `image_transport`, messages, `tf2_ros`, ...
`isaac-ros-5`, `depthai-deps`	optional: Isaac ROS 5.0; system dependencies of depthai

Workspace dependencies are collected by `jxb sysroot rosdeps -w WS`, which runs `jxb-rosdeps` in the xbuild image. It resolves all `rosdep` keys of the workspace for the board's Ubuntu release and ROS distribution, and it **also scans launch files** for the packages they start or include:

```
LAUNCH_PKG = re.compile(r"""\b(?:package|pkg)\s*=\s*["']([A-Za-z][A-Za-z0-9_]*)["']"""
                        r"""|(?:get_package_share_directory|FindPackageShare)\(\s*["']([A-Za-z][A-Za-z0-9_
    ↪ ]*)["']""")
```

The scan was added after the shipped Kilted runtime failed with "package 'topic_tools' not found": the SAM launch files start `topic_tools`, `rplidar_ros`, `mavros` and `robot_state_publisher` and include `foxglove_bridge`, but no `package.xml` declares them. On a developer PC with a full ROS installation this goes unnoticed; on a board runtime made from declared dependencies it does not (F52, F60). GUI tools and simulators are excluded. Keys that do not resolve are an error; for a distribution built for another Ubuntu release (Lyrical on 24.04), keys are retried against the release it targets and accepted only if they map to `ros-<distro>-*` packages (F31).

4.10 Metadata and the swap

The builder writes metadata into `<sysroot>/.jxb/` (Table 8). The sysroot is assembled in `<name>.staging` and swapped in by renaming, so an interrupted build never leaves a half-written sysroot behind. Because a running dev container still sees the old, now deleted directory through its bind mount, `jxb sysroot build` recreates the container with the same workspaces (F14).

Table 8: Sysroot metadata.

File	Content	Used by
`packages.lock`	`name=version` of every package	`jxb board check`
`sources.list`	the exact apt sources	audit
`info.json`	build time, counts, alternatives, links, extras with SHA-256, fixups	runtime image label, `jxb sysroot info`
`profile.json`	the fully resolved profile	`env.sh`, `cross-colcon-build`, extras

4.11 Properties

- **Determinism.** For fixed repository contents, the output is a function of the profile, and `packages.lock` documents it; `jxb board check` compares it with a board. Repositories advance over time: the Nano had `rclcpp` 28.1.21 from an earlier sync while the sysroot had 28.1.22, and

the board's cuDNN was 9.25 while JetPack specifies 9.20 (F35). Such patch-level differences are ABI-compatible and reported as DIFF.

- **Cost.** About 8 GB of downloads once, then about one minute per rebuild; 16–19 GB per sysroot.
- **No privileges.** Neither root, nor a board, nor emulation is needed.

5 Deep dive II: the cross toolchain

5.1 The xbuild image

`docker/xbuild.Dockerfile` builds a host-tools-only image (≈3.4 GB) per profile on the Ubuntu release of the board's userland (24.04, or 26.04 for `jp721-lyrical`). Its layers are:

1. build tools and the distribution's `gcc-aarch64-linux-gnu/g++-aarch64-linux-gnu` (GCC 13 on 24.04), the default compiler of the board's Ubuntu release; `qemu-user` for optional smoke tests;
2. ROS 2 on the host (`ros-<distro>-ros-base`, the rosidl generators, `ros-dev-tools`);
3. NVIDIA's **x86_64 nvcc of exactly the board's version** (`cuda-nvcc-13-2=13.2.86-1`), held at that version;
4. build-time generators the SAM stack needs on the host: `acados_template`, casadi 3.7.2 and the x86 `t_renderer` (F20, F21);
5. a non-root user with the host's UID/GID, so files written into mounted workspaces keep their owner.

The jxb tree is *not* copied into the image but bind-mounted at `/opt/jxb`, so changes to the toolchain take effect without rebuilding the image. The CUDA layer contains two deliberate modifications:

```
RUN ... && cd /usr/local/cuda-${CUDA_DASH/-/.} \
    && ln -sfn targets/${CUDA_TARGET}/include include && ln -sfn targets/${CUDA_TARGET}/lib lib64 \
    && sed -i "1i _TARGET_DIR_ = targets/${CUDA_TARGET}" bin/nvcc.profile
ENV NVCC_CCBIN=aarch64-linux-gnu-g++
```

CMake's `FindCUDAToolkit` probes `nvcc -v` without flags and then used the *x86_64* CUDA include directory for C++-only targets (F5). Making the board's target directory nvcc's default, and the aarch64 compiler its default host compiler, turns this toolkit into one that *only* targets the board: even a plain `nvcc x.cu` produces aarch64 code. The `targets/sbsa-linux` directory of the image is replaced at run time by the sysroot's copy (Section 5.6), so device code is compiled against exactly the board's CUDA headers and libraries.

Why it matters

Configure tools so that their *defaults* are correct. Build systems probe compilers in many ways, and not every probe passes the flags a toolchain file sets.

5.2 Deriving the environment

Every board-specific value reaches the toolchain through environment variables, and all of them are derived from one place, the profile copy inside the sysroot (Listing 3). `profile_env.py` maps the profile to `JXB_PROFILE`, `ROS_DISTRO`, `JXB_CPU_FLAGS` (`-mcpu=cortex-a78ae`), `JXB_CUDA_ARCH` (87), `JXB_CUDA_VERSION` (13.2), `JXB_CUDA_TARGET` (sbsa-linux), `JXB_PYTHON_VERSION` and `JXB_QEMU_CPU`. The container only needs `ARM64_SYSROOT`; everything else follows from it.

```
if [ -n "${ARM64_SYSROOT:-}" ] && [ -f "${ARM64_SYSROOT}/.jxb/profile.json" ]; then
    while IFS= read -r _jxb_line; do
```

```
        _jxb_name=${_jxb_line#export }; _jxb_name=${_jxb_name%%=*}
        if [ -z "${!_jxb_name:-}" ]; then eval "${_jxb_line}"; fi     # variables already set win
    done < <(python3 "${JXB_HOME:-/opt/jxb}/toolchain/profile_env.py" "${ARM64_SYSROOT}/.jxb/profile.json")
    ↪
    export JXB_TOOLCHAIN_FILE="${JXB_HOME:-/opt/jxb}/toolchain/orin.toolchain.cmake"
fi
```

Listing 3: `toolchain/env.sh`, sourced by every shell in the container.

5.3 The CMake toolchain file

`toolchain/orin.toolchain.cmake` is the core of the cross build [10]. It refuses to run without the environment above and then configures five areas.

Compilers and search roots.

```
set(CMAKE_SYSTEM_NAME Linux)
set(CMAKE_SYSTEM_PROCESSOR aarch64)
set(CMAKE_SYSROOT "$ENV{ARM64_SYSROOT}")
set(CMAKE_C_COMPILER aarch64-linux-gnu-gcc)
set(CMAKE_CXX_COMPILER aarch64-linux-gnu-g++)
# Append, never overwrite: toolchain files are read several times, and wrappers that
# chainload this file (vcpkg) add their own roots after the first read.
list(APPEND CMAKE_FIND_ROOT_PATH "${JXB_SYSROOT}")
if(DEFINED ENV{JXB_WS_INSTALL})                  # packages of the same colcon workspace
  string(REPLACE ":" ";" _jxb_ws "$ENV{JXB_WS_INSTALL}")
  list(APPEND CMAKE_FIND_ROOT_PATH ${_jxb_ws})
endif()
set(CMAKE_FIND_PACKAGE_RESOLVE_SYMLINKS ON)
set(CMAKE_FIND_ROOT_PATH_MODE_PROGRAM NEVER)     # programs: host tools
set(CMAKE_FIND_ROOT_PATH_MODE_LIBRARY ONLY)      # libraries, headers, packages: target only
set(CMAKE_FIND_ROOT_PATH_MODE_INCLUDE ONLY)
set(CMAKE_FIND_ROOT_PATH_MODE_PACKAGE ONLY)
```

`CMAKE_FIND_ROOT_PATH` re-roots every search into the sysroot. Packages built earlier in the same colcon run are installed *outside* the sysroot, so the workspace's install space is added as a second root. Two subtle bugs shaped this block: the first version used `set()`, which lost vcpkg's roots on the second read of the file (F33); and in a merged-`/usr` sysroot, a CMake configuration found through the link `lib` → `usr/lib` computed its install prefix as the sysroot root, which `CMAKE_FIND_PACKAGE_RESOLVE_SYMLINKS` prevents (F39).

Linker paths. The board's dynamic loader finds transitive dependencies through `/etc/ld.so.conf.d` and `LD_LIBRARY_PATH`; the cross linker reads neither. The toolchain therefore collects the sysroot's `ld.so.conf.d` entries plus the directories that are only on the board's `LD_LIBRARY_PATH` (ROS, ONNX Runtime) and passes each as `-Wl,-rpath-link`. `--sysroot` is repeated in the linker flags because CMake passes no sysroot flag when the host compiler links CUDA targets (F6).

pkg-config. `PKG_CONFIG_LIBDIR` lists only the sysroot's `.pc` directories and `PKG_CONFIG_SYSROOT_DIR` re-roots their paths; `PKG_CONFIG_PATH` is cleared so that no host `.pc` file can leak in.

Python. The host interpreter runs the generators, but headers, `libpython` and the extension suffix must be the board's. Without overriding the SOABI, extension modules would be named `*.cpython-312-x86_64-linux-gnu.so` and would not import on the board:

```
foreach(_jxb_p Python3 Python)
  set(${_jxb_p}_EXECUTABLE /usr/bin/python3 CACHE FILEPATH "Host Python for code generators")
  set(${_jxb_p}_INCLUDE_DIR "${JXB_SYSROOT}/usr/include/python${_jxb_py}" CACHE PATH "")
```

```
  set(${_jxb_p}_LIBRARY "${JXB_SYSROOT}/usr/lib/aarch64-linux-gnu/libpython${_jxb_py}.so" CACHE FILEPATH "
    ↪ ")
  set(${_jxb_p}_SOABI "cpython-${_jxb_pynodot}-aarch64-linux-gnu" CACHE STRING "")
endforeach()
set(PYTHON_MODULE_EXTENSION ".cpython-${_jxb_pynodot}-aarch64-linux-gnu.so" CACHE STRING "")  # pybind11
```

CUDA.

```
set(CMAKE_CUDA_COMPILER "/usr/local/cuda-$ENV{JXB_CUDA_VERSION}/bin/nvcc")
set(CMAKE_CUDA_HOST_COMPILER aarch64-linux-gnu-g++)
if(NOT DEFINED CMAKE_CUDA_ARCHITECTURES)
  set(CMAKE_CUDA_ARCHITECTURES "$ENV{JXB_CUDA_ARCH}")                # 87
endif()
set(CMAKE_CUDA_FLAGS_INIT
  "--target-directory $ENV{JXB_CUDA_TARGET} -Xcompiler=--sysroot=${JXB_SYSROOT},${_jxb_cpu_xc}")
set(ENV{CUDA_TOOLKIT_ROOT} "${JXB_CUDA_ROOT}")                       # legacy FindCUDA (F31)
```

The last line exists because the legacy `FindCUDA` module, still used by `isaac_ros_common`, takes the cross target directory only from an environment variable (F31).

5.4 What runs where

Table 9 lists which machine each step of a build is for; this distinction is the key to reasoning about any cross-build error.

Table 9: Where the steps of a cross build run and what they produce.

Step	Program runs on	Reads	Output for
rosidl / ament generators	host (x86 Python, host ROS)	`.msg` files, templates	source code
C/C++ compilation	host (`aarch64-linux-gnu-g++`)	sysroot headers	board
CUDA device code	host (x86 `nvcc`)	sysroot `targets/sbsa-linux`	board GPU (`sm_87`)
CUDA host code	host (aarch64 GCC via `-ccbin`)	sysroot headers	board
linking	host (`aarch64-linux-gnu-ld`)	sysroot libraries, `-rpath-link`	board
acados solver generation	host (casadi, x86 `t_renderer`)	board headers	C code for the board
TensorRT engine build	**board**	ONNX model	board (device- and version-specific)

5.5 `cross-colcon-build`

The wrapper that `jxb build` runs in the container

1. sources `env.sh` and refuses to run without a `jxb` sysroot;
2. partitions its arguments into colcon and CMake arguments (CMake's own long options such as `--fresh` stay CMake's) and forbids overriding `CMAKE_TOOLCHAIN_FILE`;
3. adds defaults unless given: `CMAKE_BUILD_TYPE=Release`, `BUILD_TESTING=OFF`, `CMAKE_EXPORT_COMPILE_COMMANDS=ON`;
4. uses per-profile trees `build-<profile>/` and `install-<profile>/`, so one workspace can be built for several distributions side by side;

5. sources `ros-cross-env.sh`: host ROS Python modules first and the sysroot's (pure Python) as fallback, `AMENT_PREFIX_PATH` and `CMAKE_PREFIX_PATH` at the sysroot's `/opt/ros/<distro>`, `CC`/`CXX` set to the board compiler for build steps that compile without CMake (acados' generated Makefiles), and `JXB_CROSS_BUILD=1`;
6. after the build, rewrites build-time paths and repairs ELF files (Listing 4).

```
# colcon's setup scripts and ament's parent_prefix_path index record the build-time underlay, i.e. the
# sysroot's copy of /opt/ros/<distro>. On the board that is plain /opt/ros/<distro>.
{ find "${install_base}" -maxdepth 1 -type f -name 'setup.*'
  find "${install_base}" -type f -path '*/share/ament_index/resource_index/parent_prefix_path/*'
} | xargs -r sed -i "s#${ARM64_SYSROOT}/#/#g"
"$(dirname "$(realpath "$0")")/jxb-fix-needed" "${install_base}"
```

Listing 4: Post-processing of the install space in `cross-colcon-build`.

5.6 The dev container

`jxb up` creates a long-lived container `jxb-<profile>` from the xbuild image with the mounts of Table 10. The board paths exist because SAM's `object_detection_package` hard-codes `/opt/onnxruntime`, a board path (F10). Mounting workspaces at their host paths means that the absolute paths in `compile_commands.json` and in the DWARF debug information are valid on the host, so clangd and gdb work there without path maps.

Table 10: Mounts of the dev container.

Mount	Source	Target	Mode
jxb tools	repository	`/opt/jxb` and its own path	ro
sysroot	`sysroots/<profile>`	same path	ro
CUDA target	`<sysroot>/usr/local/cuda-13.2/targets/sbsa-linux`	`/usr/local/cuda-13.2/targets/sbsa-linux`, `.../aarch64-linux`	ro
board paths	`<sysroot>/opt/onnxruntime`, `<sysroot>/opt/acados`	`/opt/onnxruntime`, `/opt/acados`	ro
workspaces	each workspace	**same path**	rw
ccache	volume `jxb-ccache-<profile>`	`~/.ccache`	rw

5.7 Verifying binaries

Two tools inspect every cross-built ELF file.

`jxb-elfcheck` parses `readelf` output and checks for each file that the machine type is AArch64, the interpreter is `/lib/ld-linux-aarch64.so.1`, no `RUNPATH`/`RPATH` entry contains a host path, every `DT_NEEDED` entry resolves the way it will on the board (`RUNPATH`, the install tree's library directories, the sysroot's linker paths), no `DT_NEEDED` entry is an absolute path, and every required symbol version (`GLIBC_*`, `GLIBCXX_*`, ...) is defined by the library it resolves to. The last check catches objects compiled against the cross compiler's own libc or libstdc++ instead of the board's (Ubuntu's cross GCC searches its own directories before the sysroot, F7).

`jxb-fix-needed` repairs a defect that the linker produces legitimately. A shared library *without* `SONAME` that is linked by its full path is recorded by that path, here `<sysroot>/opt/ros/lyrical/lib/libtensor_rt_node.so` (Isaac ROS exports its node libraries this way, F44). The fix exploits a property of the ELF dynamic string table: the file name is a *suffix* of the recorded path, and strings are NUL-terminated, so pointing the `DT_NEEDED` entry at the offset of the suffix yields the file name without moving a single byte (Fig. 5):

$$\text{off}' = \text{off} + |\text{path}| - |\text{basename}(\text{path})|. \tag{2}$$

The tool walks PT_DYNAMIC, translates virtual addresses to file offsets through the PT_LOAD segments, rewrites the matching DT_NEEDED entries and the vn_file fields of .gnu.version_r that name the same library (the loader matches version requirements against the name by which it loaded a library), and writes the file in place. Neither patchelf nor any section resizing is needed.

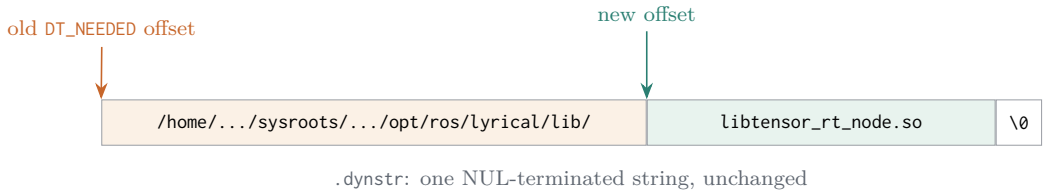


Figure 5: Rewriting an absolute DT_NEEDED entry by moving its string offset to the file-name suffix.

5.8 The test suite

jxb test cross-builds and checks probe projects that cover every layer: CUDA with relocatable device code and cuBLAS (cuda_hello), a TensorRT engine build and inference (trt_smoke), ONNX Runtime with the CPU, CUDA and TensorRT EPs (ort_smoke, ort_bench), the ONNX library built from source (onnx_lib), a ROS 2 workspace with a custom interface and a CUDA, TensorRT and cv_bridge node (ros_ws) and, for Isaac profiles, a node that uses Lyrical's rosidl::Buffer CUDA backend (isaac_ws). On the host the stage is checked with jxb-elfcheck, the extension-suffix check and runs under qemu-aarch64 -cpu cortex-a76, the closest QEMU model to the Cortex-A78AE (QEMU 10's default model raises SIGILL on such code, F25). jxb test --board B copies the stage to ~/jxb/<profile>/tests and runs it on the GPU. The parsers of the trace analysis have their own unit tests.

6 Deep dive III: deployment and the board runtime

6.1 The board model

Boards are listed in boards.toml (local, not committed) with host, user, optional SSH port, deployment directory and tool overrides. All SSH calls use keys only (BatchMode=yes; a failed login prints the ssh-copy-id command), ConnectTimeout=8 and StrictHostKeyChecking=accept-new. Everything jxb writes lives in one directory per profile (Listing 5).

```
~/jxb/
|-- .measure.lock, .measure.lock.info    measurement lock (flock) and its current holder
|-- trt_cache/                           TensorRT EP engine cache of the benchmarks
`-- <profile>/                           everything deployed for this profile; deletable at any time
    |-- <ws>/                            a deployed workspace (exported as WORKSPACE_ROOT)
    |   |-- install/                     install-<profile>/ of the host
    |   |-- launch/  assets/             workspace files
    |   `-- src/<pkg>/{launch,config,params,urdf}/
    |-- runtime/
    |   |-- opt/ros/<distro>/            shipped ROS distribution (only if the board lacks it)
    |   |-- opt/onnxruntime/ opt/acados/ runtime extras ([deploy] runtime)
    |   |-- pydist/                      Python modules the board's Python lacks
    |   `-- syslibs/                     shared libraries the board lacks
    |-- tools/                           board helpers (synthetic camera, bag to CSV, ftrace, benchmark)
    |-- tests/  bench/                   test stage; benchmark binaries, models and engines
    `-- traces/                          recordings in progress (fetched to the host, then removed)
```

Listing 5: Layout of a board's ~/jxb/ directory.

6.2 Deploying a workspace

`deploy_workspace` performs, in order:

1. **Board preparation**: in container mode, the runtime image is made available on the board; natively, the ROS distribution is shipped if the board lacks it (Section 6.3).
2. **Incomplete packages.** A package that failed to build still leaves an install directory, because colcon writes `share/<pkg>/package.dsv` before building; its hooks name `local_setup.*` files that were never installed, and sourcing the workspace on the board then stops at that package (F57). Such packages are detected by checking that every `source;` hook in `package.dsv` exists; they are not deployed, and stale copies are removed.
3. **The install space**, completely (`rsync --delete`) or only the requested `--packages` together with colcon's top-level setup files, using ordered rsync filter rules.
4. **Workspace files** needed at run time but not installed: `launch/`, `assets/` and the `launch`, `config`, `params` and `urdf` folders of the packages. Files that are only **Git LFS pointers** (smaller than 1 KiB and starting with `version https://git-lfs.github.com/spec/`) are excluded, after a pointer once replaced the real model on the board (F51).
5. **Runtime extras** (`[deploy] runtime`): jxb's ONNX Runtime GPU build and acados go to `~/jxb/<profile>/runtime/opt/...`, never into the board's `/opt`.
6. **Library closure check** (Section 6.3).

6.3 Making a stock board run another ROS distribution

The board runs Jazzy; Kilted and Lyrical must run without touching `apt` (R4, R5). Two observations make this possible. First, ROS 2's setup scripts derive their prefix from their own location, so the sysroot's `/opt/ros/<distro>` is **relocatable**: copied to `~/jxb/<profile>/runtime/opt/ros/<distro>` without headers and static libraries (191 MB for Kilted, 226 MB for Lyrical with Isaac ROS), it runs from there (F44). This requires the same Ubuntu release, which is checked against the board's `VERSION_CODENAME`. Second, the remaining gaps can be found mechanically:

- **Python modules.** Lyrical takes `osrf_pycommon` from the system package `python3-osrf-pycommon`, which the board has only inside `/opt/ros/jazzy`, so `ros2 launch` failed (F61). `ship_pydist` compares the sysroot's `dist-packages` with the modules the board's Python can import and ships the difference to `runtime/pydist`, appended to `PYTHONPATH`.
- **Shared libraries.** `ensure_board_libs` computes a fix point (Algorithm 2): a script on the board prints `HAVE <file>` for every library below the deployed directories and `MISSING <soname>` for every library that `ldd` cannot resolve for any deployed ELF file; missing libraries are copied from the sysroot to `runtime/syslibs`, which is on `LD_LIBRARY_PATH`, and the check is repeated, because shipped libraries can have dependencies of their own.

In practice the board lacked no system library for Kilted or Lyrical on Ubuntu 24.04; the mechanism turns that into a checked fact instead of an assumption.

6.4 The board environment

Every remote command is prefixed by a generated shell snippet whose order matters (Listing 6). The underlay is sourced explicitly and the workspace through `local_setup.bash`, not `setup.bash`, because the underlay path recorded at build time need not exist on the board. jxb's ONNX Runtime comes first on `LD_LIBRARY_PATH`, so it wins over the board's own CPU-only `/opt/onnxruntime` (F38). `set +u` is needed because ROS setup scripts reference unset variables (F37).

```
set +u; export PYTHONPATH=${PYTHONPATH:+$PYTHONPATH:}~/jxb/P/runtime/pydist;              # native only
export JXB_ROS_SETUP=/opt/ros/D/setup.bash; [ -f $JXB_ROS_SETUP ] ||
```

Algorithm 2 Library closure on the board (`ensure_board_libs`).

Require: deployed directories D; sysroot library directories L

$S \leftarrow \emptyset$ ▷ sonames shipped to `runtime/syslibs`
for $k = 1, \ldots, 5$ **do**
  $(H, M) \leftarrow$ run on the board: libraries present below D; sonames `ldd` cannot resolve
  $M \leftarrow M \setminus (H \cup S)$
  if $M = \emptyset$ **then return** ▷ every deployed ELF file resolves
  end if
  for $m \in M$ **do**
    if $\exists \ell \in L : \ell/m$ exists in the sysroot **then** copy it to the stage; $S \leftarrow S \cup \{m\}$
    else fail: "missing on the board and not in the sysroot"
    end if
  end for
  rsync the stage to `runtime/syslibs`
end for
fail: no fix point after five rounds

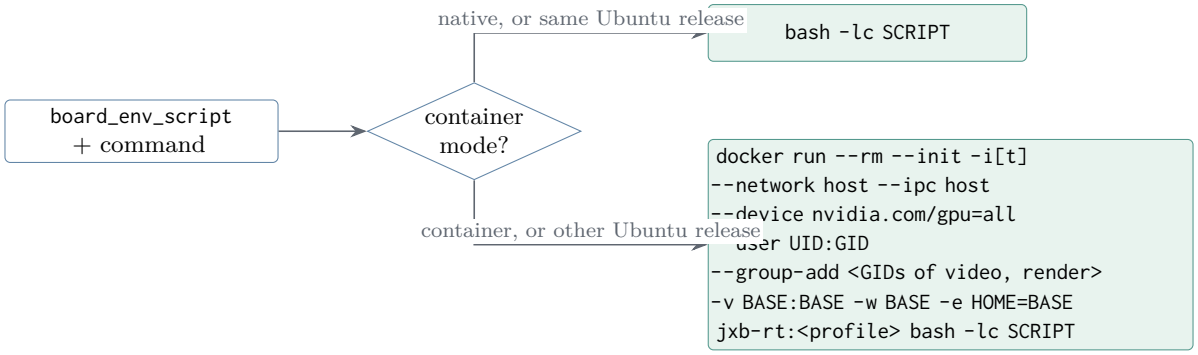


Figure 6: Native and container execution of the same script.

```
    JXB_ROS_SETUP=~/jxb/P/runtime/opt/ros/D/setup.bash; source $JXB_ROS_SETUP &&          # 1. underlay
if [ -f ~/jxb/P/WS/install/local_setup.bash ]; then source ~/jxb/P/WS/install/local_setup.bash; fi &&  # 2.
    ↪
export LD_LIBRARY_PATH=~/jxb/P/runtime/opt/onnxruntime/lib:~/jxb/P/runtime/opt/acados/lib:~/jxb/P/runtime/
    ↪ syslibs:$LD_LIBRARY_PATH &&
export WORKSPACE_ROOT=~/jxb/P/WS && export ROS_AUTOMATIC_DISCOVERY_RANGE=LOCALHOST && ...  # 3. jxb.toml
    ↪ env
```

Listing 6: The environment prefix of a remote command (`board_env_script`; abridged, `P` = profile, `D` = distribution, `WS` = workspace).

6.5 Native and container mode

`uses_container` decides where board commands run: an explicit `--mode native|container` (or `mode` in `jxb.toml`) wins; otherwise a profile can force a mode (`[deploy] container`); otherwise the container is used exactly when the profile's Ubuntu release differs from the board's. `board_exec` then wraps the script (Fig. 6).

The **runtime image** is created on the host, without emulation, by streaming the sysroot into `docker import`:

```
tar = subprocess.Popen(["tar", "-C", str(p.sysroot), "-c", "--numeric-owner",
                        *(f"--exclude={x}" for x in RUNTIME_IMAGE_EXCLUDES), "."], stdout=subprocess.PIPE)
sh(["docker", "import", "--platform", "linux/arm64", "--change", f"LABEL jxb.profile={p.name}",
    "--change", f"LABEL jxb.sysroot.built={built}", "--change", 'CMD ["bash"]', "-", tag], stdin=tar.
    ↪ stdout)
```

The excludes drop headers, static libraries, documentation, the CUDA compiler and the TensorRT builder resources for GPU architectures that Orin's TensorRT does not load, which yields images of 10.2–10.8 GB. The label records the sysroot's build time, and the board's copy is replaced only when the label differs, by streaming `docker save | zstd -T0 -3 | ssh 'zstd -d | docker load'` and removing the previous image. The flags of `docker run` encode several findings (Table 11).

Table 11: Flags of the runtime container and their reasons.

Flag	Reason
`--device nvidia.com/gpu=all`	the GPU through the board's CDI specification [14]; JetPack's container toolkit mounts the host's CUDA driver and libraries
`--group-add` GIDs of `video`, `render`	GPU device nodes belong to these groups; numeric, since the image (a sysroot) has no such group names (F48)
`--network host --ipc host`	DDS discovery and shared-memory transport with nodes outside the container
`-v BASE:BASE -w BASE`	the deployment at the same path as natively: the same scripts work in both modes
`--user $(id -u):$(id -g)`, `-e HOME=BASE`	files written in the container belong to the board user; `HOME` is writable
`--init`, `-t` with a terminal	signals and hang-ups reach the processes; closing the connection stops the container
optional `-v /usr/bin/gdbserver`, `/opt/nvidia/nsight-systems`	the board's debugger and profiler inside the container

Why a 26.04 userland runs on a 24.04 board

The Ubuntu 26.04 userland of `jp721-lyrical` runs with the CUDA driver and libraries of the 24.04 host. This is sound because the container toolkit mounts the host's driver stack, and newer glibc versions remain backward compatible with libraries built for older ones. NVIDIA publishes neither CUDA 13.2 nor TensorRT for Ubuntu 26.04 on SBSA (F23).

6.6 Launch mode

`jxb launch BOARD [options] -- TARGET [ARGS]` is the Renesas-style remote launch: deploy, start a launch file on the board, stop it cleanly. `TARGET` is either `PKG FILE` or a launch file of the workspace (for SAM: `launch/launch.py module:=parking`), which is mapped to its deployed copy. The command composes three layers of shell (Fig. 7).

The difficult part is **lifecycle management**. A remote launch must stop completely on Ctrl-C, at the end of `--duration`, on a lost connection and when `ros2 launch` itself dies, and it must never stop processes that do not belong to it. Earlier versions failed in instructive ways (F58, F62): nodes of a launch that died early kept running and blocked `nsys` for 15 minutes; a sweep with `pkill -f <deploy path>` also killed jxb's own wrapper shells, whose command lines contain that path; and a leftover `tegrastats` held the SSH channel open. The final design relies on POSIX process groups (Listing 7).

```
# helpers, the recorder and the launch each in a session of their own; $! is the group id
setsid bash -c "$HELPER" > with0.log 2>&1 & groups="$groups $!"
setsid ros2 bag record -s mcap -o bag TOPICS > bag.log 2>&1 & JXB_BAG=$!
setsid bash -c 'ros2 launch ... 2>&1 | tee -i launch.log; exit ${PIPESTATUS[0]}' & JXB_L=$!
echo $JXB_L > pgid
trap 'kill -INT -- -$JXB_L 2>/dev/null' INT TERM HUP     # forward Ctrl-C and hang-ups to the launch
wait $JXB_L; rc=$?; while kill -0 $JXB_L 2>/dev/null; do wait $JXB_L; rc=$?; done
trap - INT TERM HUP
for g in $groups; do kill -TERM -- -$g 2>/dev/null; done   # helpers: SIGTERM
```

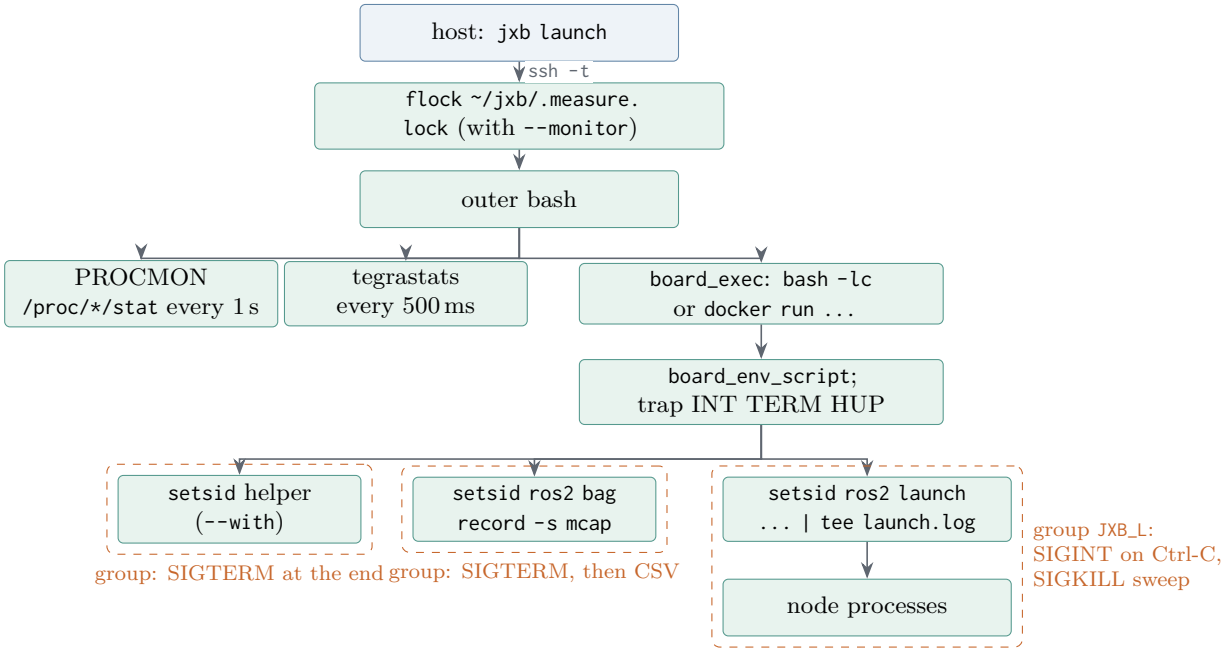


Figure 7: Process structure of `jxb launch --monitor --bag --with ...`. Dashed boxes are process groups created with `setsid`; the monitors run on the board host, so they also see the processes of a runtime container.

```
kill -TERM -- -$JXB_BAG; wait $JXB_BAG; python3 tools/jxb_bag_csv.py bag   # close the bag, numeric topics
    ↪ as CSV
sleep 1; for g in $JXB_L $groups; do kill -KILL -- -$g 2>/dev/null; done  # sweep leftovers
touch .done; exit $rc
```

Listing 7: The launch script (abridged from `toolchain/jxb_launch.py`).

Three POSIX details are essential. `setsid` does not fork for a background job of a non-interactive shell, so `$!` is both the PID and the ID of the new process group, and `kill -- -PGID` reaches every process of that tree, including nodes whose `ros2 launch` parent has died. Background jobs of a non-interactive shell ignore SIGINT, which is why helpers are stopped with SIGTERM. And `wait` returns early when a trapped signal arrives, hence the loop. Without a terminal (VS Code, scripts), a local Ctrl-C cannot reach the board's process group through SSH; `jxb` catches the `KeyboardInterrupt`, sends SIGINT to the recorded group (through `docker exec` in container mode, since the PID belongs to the container's PID namespace) and waits for `.done` before it fetches the recording. Exit codes 0, 124 (timeout), 130 and −2 (SIGINT) count as a normal end.

`--with` starts helpers in the same environment and mode, for example the synthetic camera `jxb_image_pub.py`, which renders a road scene with lane markings and moving objects and publishes `sensor_msgs/Image`, optionally with a matching `CameraInfo`, at a fixed rate, so that perception nodes can be exercised without hardware. `--bag` records MCAP; afterwards `jxb_bag_csv.py` exports numeric topics and header-stamp latencies as CSV for the analysis.

6.7 Remote debugging

`jxb debug BOARD --pkg P --exe E` starts `gdbserver --once 127.0.0.1:PORT` on the board and reaches it through a local port forward of the same SSH connection, so nothing listens on the lab network and no firewall rule is needed (Fig. 8). gdb is configured with `set sysroot <sysroot>` (target libraries with symbols), `solib-search-path` over the workspace's library directories and the runtime extras, `debuginfod` off and pending breakpoints on. The host's gdb is used if it supports aarch64 (Arch-based distributions build it with `--enable-targets=all`), otherwise `gdb-multiarch` in the container. `ssh -tt` gives the remote side a terminal, so closing the connection hangs up

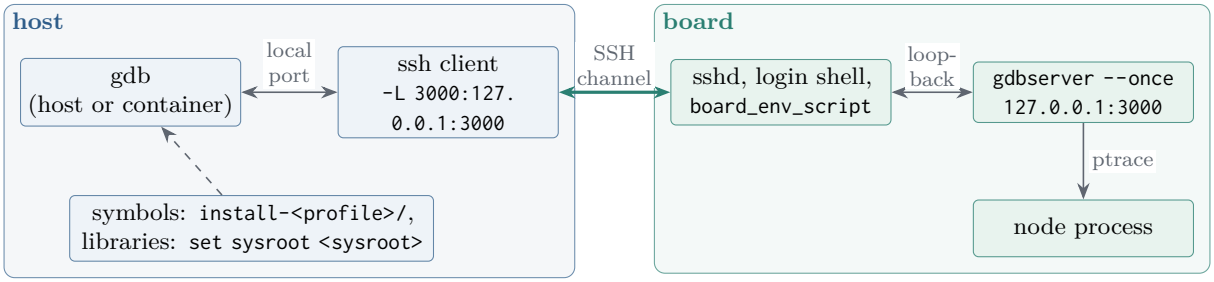


Figure 8: SSH-tunnelled remote debugging: gdbserver listens only on the board's loopback interface.

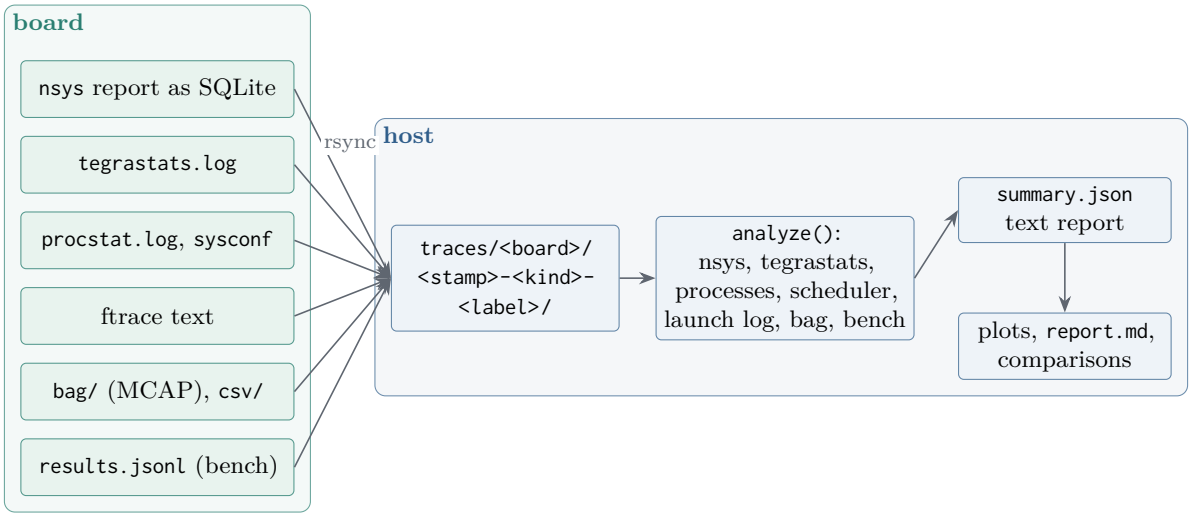


Figure 9: From raw recordings to reports.

everything it started (F42). With `--launch PKG FILE` the node is started by its launch file and only the selected executable runs under gdbserver, via `ros2 launch ... --launch-prefix 'gdbserver ...' --launch-prefix-filter '^E$'`. Attaching to a running node is not offered: the board has `kernel.yama.ptrace_scope=1`, which would require root.

7 Deep dive IV: observability

7.1 Recording model

Every measurement produces a directory `traces/<board>/<stamp>-<kind>[-<label>]/` on the host with the raw data, a `meta.json` (board, profile, command, start, wall time, exit code) and a `summary.json` computed by `analyze()` (Fig. 9). Stamps come from the host clock, because boards may run without NTP (the Nano's clock was at 1970, F35). Raw data is written on the board below `~/jxb/<profile>/traces/` (the only writable mount of the runtime container), fetched with `rsync` and removed from the board.

Measurement lock. On a shared board, a measurement must not include another user's load (F53). Every recording command is wrapped in an advisory lock (Listing 8). `flock -o` closes the lock's file descriptor before running the command, so children that outlive it cannot hold the lock; `-E 75` gives a distinct exit code on time-out. Any other tool can take part with `flock ~/jxb/.measure.lock <command>`; the hardware-in-the-loop session that shares the Nano does so.

```
held = f"echo {q(note)} > {q(lock)}.info; {command}; rc=$?; rm -f {q(lock)}.info; exit $rc"
```

```
return (f"flock -n {q(lock)} true 2>/dev/null || echo \"[jxb] {board.name} is busy with another
    ↪ measurement "
        f"($(cat {q(lock)}.info 2>/dev/null || echo 'holder unknown')); waiting up to {timeout} s\"; "
        f"exec flock -o -w {timeout} -E 75 {q(lock)} bash -c {q(held)}")
```

Listing 8: The measurement lock (`locked()` in `jxb_trace.py`).

7.2 Nsight Systems

`jxb trace nsys BOARD -- CMD` runs `nsys profile --trace cuda,nvtx,osrt,cublas,cudnn` with process-tree CPU sampling and context-switch tracing [8], which works without root on a stock board (`perf_event_paranoid=2`; system-wide sampling does not, `F45`). `--kill 2` ends the profiled process with SIGINT, so ROS nodes shut down cleanly. The report is exported to SQLite on the board (`nsys export --type sqlite`), and `nsys_summary` evaluates it with SQL over the CUPTI tables, for example:

```
SELECT s.value, count(*), sum(k.end - k.start), max(k.end - k.start)
FROM CUPTI_ACTIVITY_KIND_KERNEL k JOIN StringIds s ON s.id = k.shortName
GROUP BY k.shortName ORDER BY 3 DESC
```

The sum of kernel durations overstates the GPU utilization when kernels of several streams overlap. The **GPU busy time** is therefore the measure of the union of the kernel intervals (Fig. 10):

$$B = \Big|\bigcup_i [s_i, e_i)\Big|, \qquad u_{\mathrm{GPU}} = \frac{B}{t_1 - t_0}, \tag{3}$$

computed in one pass over the intervals sorted by start time: an interval that starts after the current merged interval ends closes it, otherwise it extends it. $[t_0, t_1]$ is the span covered by kernels, CUDA API calls, OS runtime calls and NVTX ranges. The summary further contains memory copies by direction, CUDA API calls, NVTX ranges, OS runtime calls and the hottest leaf functions of the CPU samples. Sections whose tables differ between nsys versions fail individually and are listed in `errors`, while the other sections remain.

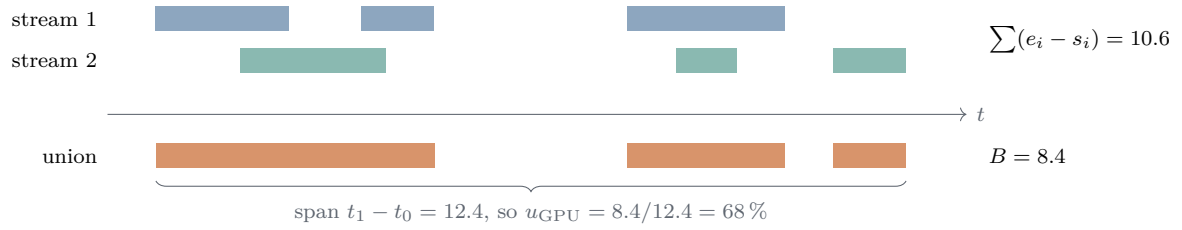


Figure 10: GPU busy time as the measure of the union of kernel intervals; the plain sum counts overlaps twice.

7.3 tegrastats

`tegrastats` is JetPack's telemetry tool and needs no root. A line such as the following (abridged, from the stack recording in Section 9.5; note the board's clock at 1970)

```
01-15-1970 00:39:37 RAM 1168/7486MB (lfb 17x4MB) CPU [40%@1344,2%@1344,38%@1344,20%@1344,0%@1344,1%@1344]
GR3D_FREQ 0% cpu@49C/49C ... gpu@48.187C/48.25C tj@49C/49C ... VDD_IN 5811mW/5816mW/5821mW VDD_CPU_GPU_CV
    ↪ 1669mW/...
```

is parsed by regular expressions into a flat record (memory, swap, load and frequency per core, GPU load and frequency, memory-controller load, temperature per zone, power per rail), written as CSV and summarized as mean, minimum and maximum per metric. With the sampling interval

Δt and the input power samples P_k of rail VDD_IN, the energy of a recording is

$$E = \sum_k P_k \, \Delta t. \tag{4}$$

tegrastats runs on the board host, also in container mode, without inherited standard streams (< /dev/null > /dev/null 2>&1), so a leftover can never hold the SSH channel open, and it is stopped by an exit trap (F62).

7.4 Per-process monitoring

jxb launch --monitor samples every process of the board once per second (Listing 9). Each T line starts a sample; the /proc/<pid>/stat lines give the parent PID, the user and system CPU ticks (utime + stime), the start time and the resident set size. Attribution is by **ancestry**: R is the PID of the session's shell and C the PID of the init process of its runtime container, and a process belongs to the session if one of its ancestors is such a root. This is robust against processes of other users that start during the recording (F53).

```
echo "R $$"; while :; do echo "T $(cut -d" " -f1 /proc/uptime)"
  if [ -n "$JXB_CN" ] && [ -z "$JXB_CP" ]; then
    JXB_CP=$(docker inspect -f "{{.State.Pid}}" "$JXB_CN" 2>/dev/null | grep -v "^0$") && echo "C $JXB_CP";
    ↪ fi
  cat /proc/[0-9]*/stat 2>/dev/null; sleep 1; done
```

Listing 9: The per-process sampler (PROCMON).

For a process with tick counts k_1, k_2 at the sample times t_1, t_2 and the tick rate h (CLK_TCK), the load and the share of a recording of length T are

$$\mathrm{cpu}(t_2) = 100\,\frac{k_2 - k_1}{h\,(t_2 - t_1)}\ \%, \qquad \mathrm{share} = 100\,\frac{k_{\mathrm{last}} - k_{\mathrm{first}}}{h\,T}\ \%, \tag{5}$$

where 100 % is one core. The share divides by the whole recording, so short-lived processes do not inflate the total, whereas the mean divides by the process's lifetime. The sampler's own cost is reported separately (monitor_cpu_pct, about 1.1 % of one core).

7.5 Kernel tracing

jxb trace kernel uses plain ftrace via tracefs [13], which every stock kernel provides. The board helper jxb-ftrace.sh saves the tracing state (tracing_on, buffer size, tracer, enabled events, record-tgid), enables the requested event groups (scheduler, IRQ, power, system calls, work queues), records or streams, and **restores the previous state on every exit**, since the board is shared. Root is required (tracefs is drwx------): by default sudo asks the user for the password in the terminal (jxb never handles passwords); alternatives are a privileged container (--root docker) or none.

SchedStats turns the ftrace text into scheduler statistics in one pass: run time per task (time between being switched in and out on a CPU), busy time per CPU, context switches and preemptions (prev_state=R), IRQ and softirq handler times, and the **wakeup latency** of every task,

$$L = t_{\text{switch-in}} - t_{\text{wakeup}}, \tag{6}$$

reported as p50, p99 and maximum per task. On the board's PREEMPT_RT kernel this is the quantity that determines the reaction time of control nodes. The same class drives a live view with per-second statistics.

7.6 Inference benchmarks

`jxb trace bench BOARD --models M.onnx ...` runs a matrix of backends on the board: ONNX Runtime with the CPU, CUDA and TensorRT EPs (fp32, fp16 and, for models with Q/DQ nodes, INT8) through jxb's `ort_bench`, and native TensorRT engines through `trtexec` (fp32, fp16, INT8). Each run yields the p50, p90 and p99 latency, the GPU compute time (trtexec) and the throughput as one JSON line. The summary sorts the backends by p50 and reports the speed-up against ONNX Runtime's CUDA EP, the "no TensorRT" baseline,

$$S = \tilde{t}_{\text{ort-cuda}} \,/\, \tilde{t}_{\text{backend}}, \tag{7}$$

where $\tilde{t}$ is the median latency. Engines are built on the board, because they are specific to the GPU and the TensorRT version, and kept for reuse, for example as the SAM node's `model:=<...>.engine`.

7.7 Reports, plots and comparisons

`jxb trace report` prints the summary of a recording, `jxb trace plot` renders plots and a Markdown report with matplotlib in the dev container, and `jxb trace compare A B ... --name N` puts several recordings side by side in `traces/compare/N/`. The figures of Section 9.5 were produced this way.

8 VS Code integration

Like the Renesas RDK [3], jxb integrates with VS Code through tasks and debug configurations, but every task only calls jxb, so the terminal and the editor run the same code (R6). `jxb vscode init -w WS` writes a multi-root workspace file `vscode/<ws>.code-workspace` (the ROS workspace plus the jxb repository); the workspace's own `.vscode/` stays untouched. What the tasks act on is set in the workspace settings, which take the role of the RDK's `settings.json` variables: `jxb.profile`, `jxb.mode`, `jxb.board`, `jxb.package`, `jxb.executable`, `jxb.launchPackage`, `jxb.launchFile`, `jxb.launchArgs`, `jxb.launchWith`, `jxb.bagTopics`, `jxb.benchModels`, `jxb.gdbPort` and `jxb.traceSeconds`. Tasks reference them as `${config:jxb.board}`; running `init` again regenerates tasks and configurations and keeps the settings (Table 12).

Table 12: Generated VS Code tasks.

Group	Tasks (all prefixed "jxb:")
build	build (default build task, `$gcc` problem matcher), build package (debug info), ... + deploy
run	deploy, run node on board, launch on board, launch + monitor (recorded), launch + monitor + bag (recorded), ssh to board
debug	gdbserver node, gdbserver node in launch, stop gdbserver launch
trace	trace nsys node, trace nsys launch, tegrastats live, kernel trace, kernel trace live (top), trace report, trace plot, benchmark models
checks	board check, test suite on board, container shell

The two debug configurations use `cppdbg` with `miDebuggerPath` set to `toolchain/bin/jxb-gdb` (which runs `jxb gdb` with the profile's sysroot) and `miDebuggerServerAddress` `127.0.0.1:${config:jxb.gdbPort}`. Their `preLaunchTask` starts `jxb debug --server-only` as a **background task**. VS Code must know when the server is ready; a problem matcher with a never-matching pattern whose `beginsPattern` and `endsPattern` are `Listening on port` signals exactly that. For a node in a launch file, a `postDebugTask` terminates the launch task through an input of type `command` (`workbench.action.tasks.terminate`). The flow was verified by replaying the generated configuration over gdb's machine interface the way `cppdbg` drives it (F47).

9 Evaluation

9.1 Setup

Table 13: Experimental setup.

Host	x86_64, 20 cores, 15 GB RAM, Docker, Arch-based Linux
Board	Jetson Orin Nano 8 GB, JetPack 7.2.1 (L4T 39.2.1), 25 W mode, GPU at 918 MHz, PREEMPT_RT kernel, NVMe
Workload	SAM ROS 2 workspace (snapshot 7552ca7f); detector models S0 (YOLO, fp32, $1 \times 3 \times 608 \times 960$) and quantized variants
Input	synthetic camera `jxb_image_pub.py`, 30 fps, 1280×800 for the stack

The board is shared with other users; all measurements ran under the measurement lock. Each configuration was recorded once for 40–120 s unless stated otherwise, so the numbers are indicative baselines rather than statistically established results (Section 9.6).

9.2 Build side

Table 14: Sysroots, test results and SAM workspace per profile.

Profile	Packages	Size	Tests (host / board)	SAM workspace
`jp721-jazzy`	944	16 GB	12 / 8	all but 2 packages, which need Kilted APIs (F15)
`jp721-kilted`	995	16 GB	14 / 9	all but `pylon_camera_package` (Basler SDK) and the vendored `micro_ros_agent` (F54)
`jp721-lyrical-isaac`	1068	19 GB	15 / 10	all but `localization_package` (Lyrical tf2, F28) and Pylon; `oak_camera_package` with jxb's depthai
`jp721-lyrical`	1100	17 GB	12 / container	all but `localization_package` and Pylon (F28)

The SAM workspace cross-builds with Kilted in 2:18 min cold and 1:14 min with a warm ccache (F19), and `jxb-elfcheck` reports no linkage problem. A sysroot rebuild from the cache takes about one minute. A fully parallel build initially ran out of memory on the 15 GB host: colcon runs `make -j<cores>` per package, so six parallel packages meant up to 120 compiler processes (F54); `MAKEFLAGS=-j4` with three parallel packages builds all 26 packages in 3.5 min.

9.3 Inference baselines

Table 15 and Fig. 11 give the latency of the SAM detector models per backend.

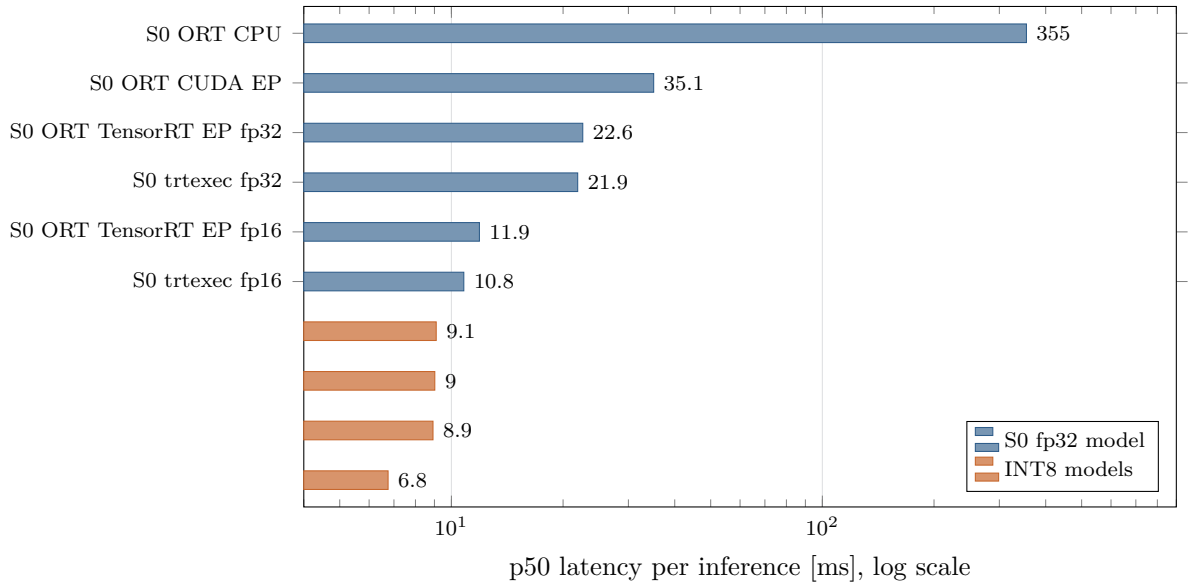


Figure 11: Median inference latency per backend (log scale).

Table 15: Inference benchmark on the Orin Nano ($1 \times 3 \times 608 \times 960$ input). Speed-up relative to ONNX Runtime's CUDA EP.

Model	Backend	p50 [ms]	p99 [ms]	Rate [1/s]	Speed-up
S0 fp32	ONNX Runtime CPU	355.0	515.1	2.5	0.10
	ONNX Runtime CUDA EP	35.1	36.1	28.5	1.00
	ONNX Runtime TensorRT EP fp32	22.6	27.3	43.8	1.56
	`trtexec` fp32	21.9	25.5	45.2	1.60
	ONNX Runtime TensorRT EP fp16	11.9	14.1	82.1	2.95
	`trtexec` fp16	**10.8**	12.0	90.9	3.24
P1 PTQ INT8	`trtexec` INT8	9.10	9.49	109.1	5.20
S2 QAT INT8	`trtexec` INT8	9.02	9.78	110.2	5.25
Q4 full INT8	`trtexec` INT8	8.92	9.19	111.4	5.36
S3 QAT INT8 (n-model)	`trtexec` INT8	**6.75**	7.18	146.6	3.98

Three observations follow. (1) TensorRT is the decisive factor: fp16 engines are 3.2× faster than ONNX Runtime's CUDA EP. (2) ONNX Runtime's TensorRT EP comes within about 10 % of `trtexec` (11.9 vs. 10.8 ms), so an application can keep the ONNX Runtime API and still obtain most of the benefit. (3) INT8 reduces the latency by only about 17 % relative to fp16 for the s-sized model; the smaller n-model is the larger lever. Without TensorRT, the Q/DQ models are *slower* than fp32 (CUDA EP 47 ms vs. 35 ms), because the quantization nodes are executed as fake quantization. An Nsight Systems trace of the CUDA EP run explains part of its cost: the inputs come from pageable host memory and are staged synchronously (`cudaMemcpyAsync` took 962 ms of API time for 430 MB that the GPU copied in 55 ms), and layout-conversion kernels take 6.4 % of the GPU time (`F46`).

9.4 The SAM detector node

In the ROS node (Fig. 12), inference takes 46.4 ms (p50) with ONNX Runtime's CUDA EP, 17.2 ms with a TensorRT fp16 engine and 15.1 ms with an INT8 (QAT) engine. The TensorRT backends also halve the session's memory (1.79 vs. 0.85 GB) and save about 1 W (6.9 vs. 5.9 W). The node adds 6–11 ms around the engine, and it runs a fixed 5 Hz loop, so the GPU is busy only 4–13 % of

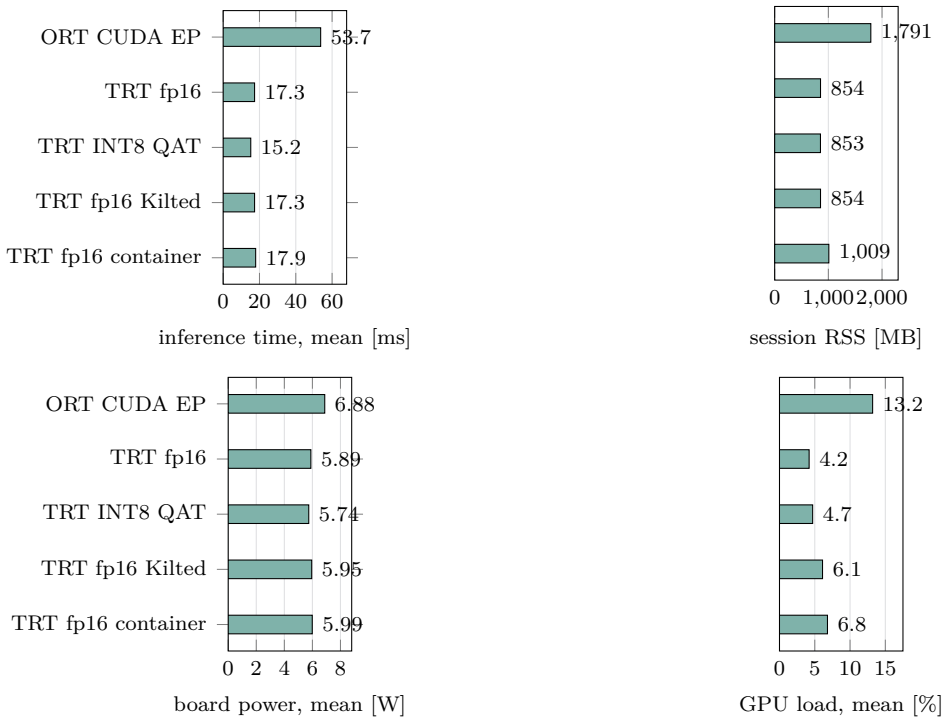


Figure 12: SAM's `object_detection` node with the synthetic camera: the first three rows are Jazzy natively, the last two Kilted natively and in the runtime container (`jxb trace compare`).

the time; the detection rate is bounded by that loop, not by the GPU (`F50`). Jazzy and Kilted give the same results within noise; under Lyrical (Isaac Buildfarm, native) memory and CPU are the same as well (1.83 GB with ONNX Runtime, 0.89 GB with TensorRT); its inference times were not recorded in that campaign, because bag recording failed with the Lyrical runtimes (fixed since, `F68`: a later run gives a median of 17.3 ms). The runtime container differs from the first native run by +0.6 ms and +0.14 GB, but a second native run differs from it by as much (+0.5 ms, +0.14 GB): the container's cost is within run-to-run variation (see the measurement report, `measurements.pdf`).

9.5 The full SAM stack

The complete stack (`launch/launch.py module:=parking`: core, object detection, parking and traffic lights; Kilted; TensorRT fp16 detector) ran for 120 s with the synthetic camera at 30 fps, natively and in the runtime container (Table 16).

Table 16: The SAM stack on the Orin Nano (120 s each). CPU in cores (1.0 = one core fully used).

Configuration	Proc.	CPU	RSS [GB]	GPU mean/max [%]	Power [W]	T_{max} [°C]
native, TensorRT fp16	12	2.83	2.11	12.6 / 84	8.36	53.0
container, TensorRT fp16	12	2.75	2.07	12.5 / 88	8.32	53.2
native, ONNX Runtime CUDA EP	12	2.93	2.92	26.8 / 99	9.39	54.5
native, one process per node	18	2.36	1.92	13.5 / 88	8.11	53.6

All processes started and stopped cleanly in every configuration. The composable-node container that hosts the image pipeline dominates (1.6 cores, 534 MB, Fig. 13); the detector takes 0.33 cores and 772 MB, the synthetic camera 0.36 cores. Lane points are produced at 23 Hz and undistorted images at 20–23 Hz from the 30 fps input, so the image pipeline does not keep up with the camera; detections arrive at the node's 5 Hz. In the loaded system, the detector's median inference time rises from 17 ms (alone) to 25 ms. An Nsight Systems trace of the stack shows the GPU busy 6.2 %

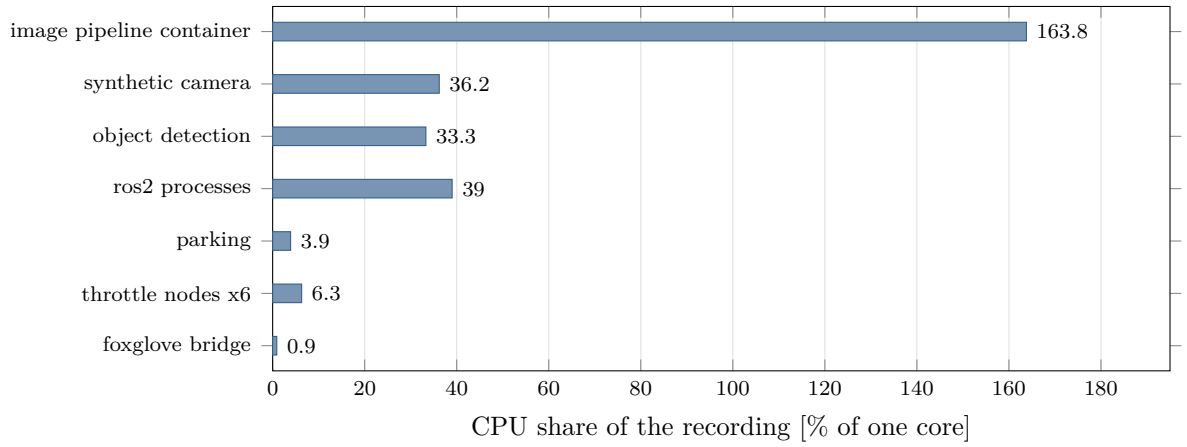


Figure 13: CPU share per process of the SAM stack (native, TensorRT fp16).

of the time, with 2.86 GB host-to-device and 3.10 GB device-to-host copies in 46 s: the pipeline moves images between CPU and GPU memory in every stage. Figures 14 and 15 show the recorded time series.

9.6 Discussion and threats to validity

The results support three conclusions about the system. First, the cross-built binaries behave like native builds: all tests pass on the board, and the SAM stack runs unchanged under three ROS distributions. Second, container mode is a viable alternative to native execution: the differences (+0.6 ms inference, +0.14 GB, ±3 % CPU) are no larger than those between two native runs. Third, the observability pipeline is cheap enough for routine use (about 1 % of a core).

The measurements have limitations. Each configuration was recorded once, on one board, which other users shared; the lock serializes measurements, but not, for example, the thermal history. The synthetic camera replaces real sensors, so the planning and control chain stays idle without odometry. Clock and thermal state were recorded but not controlled. The numbers are baselines for comparisons on this platform, not general performance claims.

10 Challenges and lessons learned

Of the 62 findings in `docs/findings.md`, those that describe problems fall into seven groups. For each group we state symptoms, root causes and resolutions, and the general lesson.

10.1 CUDA cross-compilation

- **Layout change.** CUDA 13 on Orin uses `targets/sbsa-linux` (F2); `jxb` adds the old name as an alias for older CMake versions.
- **Probing.** `FindCUDAToolkit` probes `nvcc` without flags and picked up x86 include directories for C++-only targets (F5); the image's toolkit was made target-only.
- **Missing sysroot flag.** CMake passes no `--sysroot` when the host compiler links CUDA targets (F6).
- **A compiler bug.** CUDA 13.2's CCCL headers declare specializations in a form that GCC 13/14 reject when nvcc re-emits them (F22); the fix suggested upstream is insufficient for `.cu` files, a namespace-scoped declaration works.

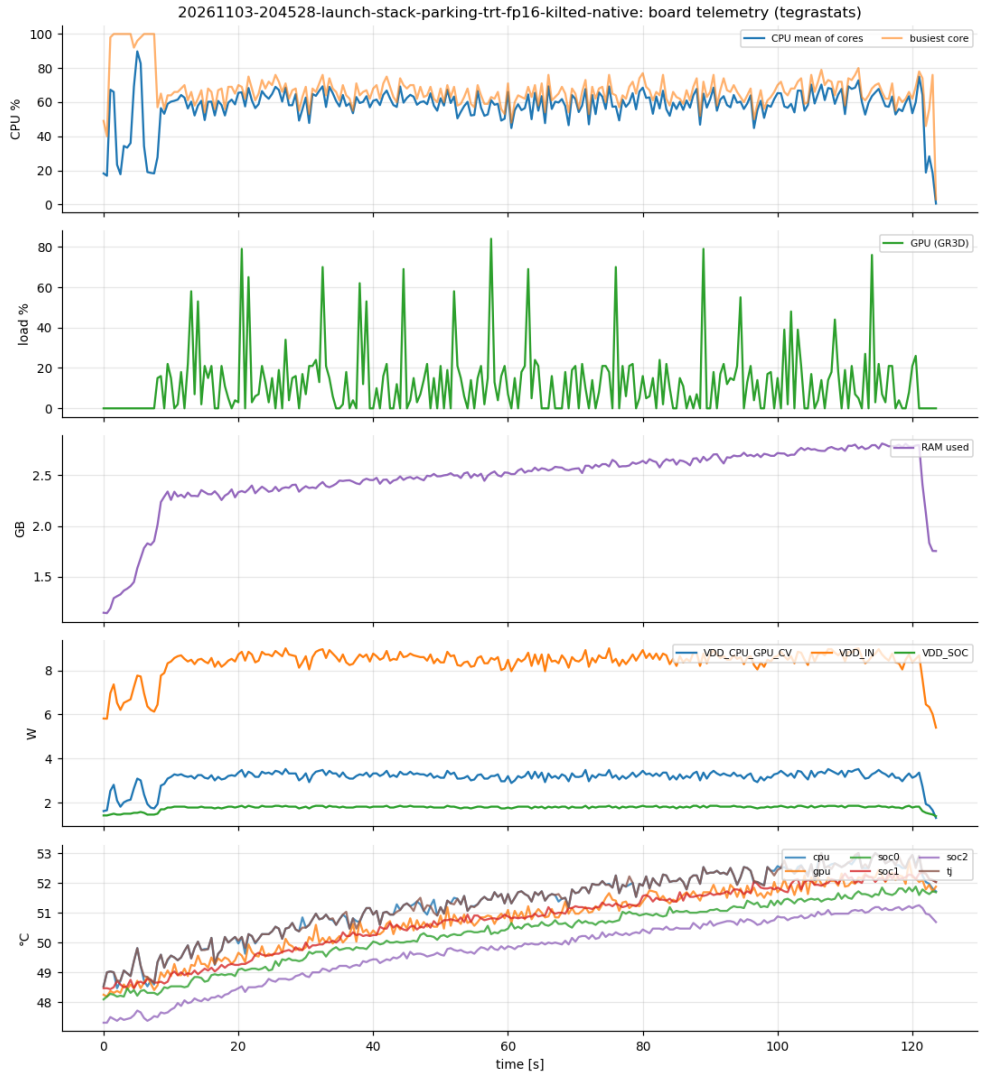


Figure 14: tegrastats of the stack run (native, TensorRT fp16): CPU, GPU, memory, power rails and temperatures over time, as rendered by `jxb trace plot`.

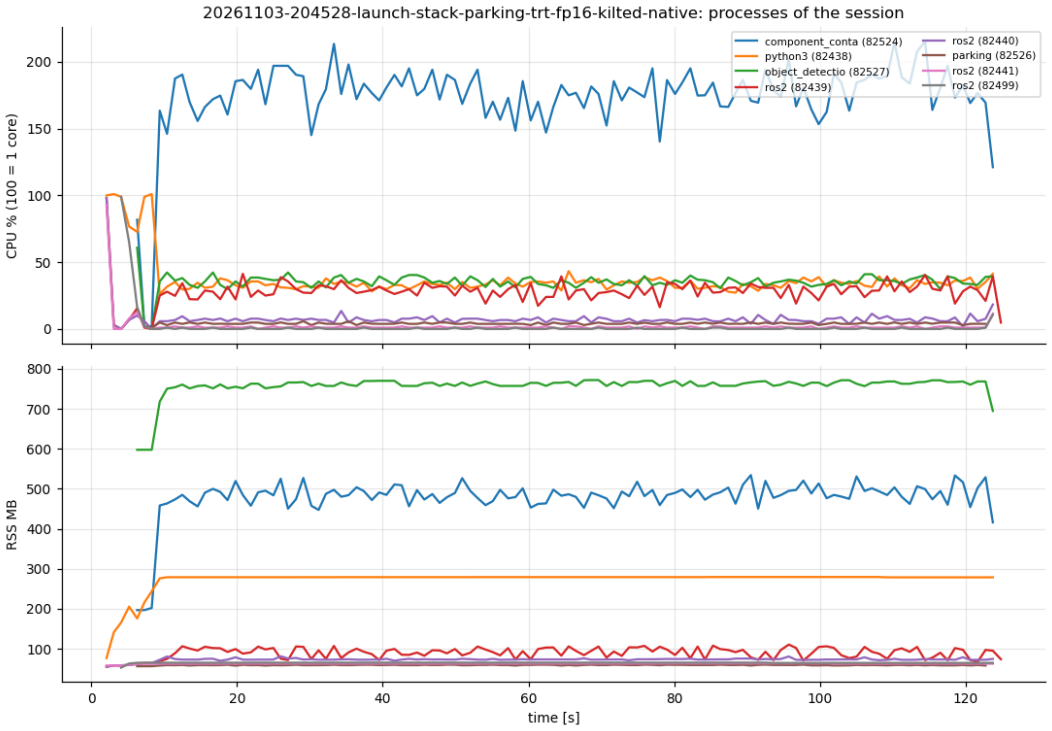


Figure 15: CPU (100 = one core) and resident memory of the stack's processes over time.

- **Legacy modules.** `FindCUDA` takes the target directory only from an environment variable (F31).

Lesson. Make the defaults of every tool correct for the target instead of relying on flags; check the target layout of every new JetPack major release.

10.2 Sysroot semantics

- Alternatives (F43), literal `ln -s` (F13) and a boot-time service (F26) create files that builds need.
- Key formats (F16) and pins (F17) must be reproduced exactly.
- Symbolic links must be relative, and CMake must resolve them before computing prefixes (F39).
- Toolchain files are read more than once (F33): use idempotent `list(APPEND)`, never `set()`.

Lesson. Enumerate the side effects of maintainer scripts instead of executing them, and skip what cannot be determined statically rather than guessing.

10.3 Binary portability

- Absolute `DT_NEEDED` entries from libraries without `SONAME` (F44) are invisible to a check that runs where the path exists.
- Build-time paths in generated files: colcon setup scripts and ament's `parent_prefix_path` record the sysroot path (F37); they are rewritten after every build.
- Python extension suffixes (F12) and CPU models for emulated tests (F25).

Lesson. Verify binaries under the target's view of the file system, not the host's.

10.4 Board runtime

- Essential packages are never dependencies, so a sysroot has no shell until they are requested (F48).
- Group names versus numbers in containers whose image is a sysroot (F48).
- Undeclared run-time dependencies in launch files (F52, F60) and system Python modules of another distribution (F61).
- Incomplete packages break sourcing a whole workspace (F57); Git LFS pointers silently replace data (F51).

Lesson. A runtime assembled from declared dependencies is a strict test of those declarations; treat its failures as bugs of the workspace, not of the runtime.

10.5 Process lifecycle on remote boards

Remote processes outlive their controllers in many ways: an early-dying launch leaves its nodes behind, `pkill -f` hits the wrong processes, and background tools inherit SSH file descriptors (F58, F62).

Lesson. Manage process groups, not processes; never match command lines; give background tools no inherited standard streams; and make cleanup part of every exit path (traps).

10.6 Shared hardware

Overlapping measurements (F53) led to the measurement lock and ancestry-based attribution. More seriously, the Nano reset about 3 s into the first run of ONNX Runtime's CUDA EP inside the runtime container, while the EP was initializing (F56). Nothing unusual preceded it (1.5 GB memory used, 49 °C, 6.7 W); TensorRT in the same container ran fine before and after; the CUDA EP runs fine natively. The container differs from native execution in its cuDNN version (9.20 from the sysroot vs. 9.25 on the board). The cause cannot be determined without root, since the panic log would be in `/sys/fs/pstore`, so this combination is excluded from container runs until then.

Lesson. On shared hardware, an unexplained fault is a policy decision as much as a bug: exclude the configuration, document it, and keep the evidence.

10.7 Upstream issues found

Table 17: Issues in upstream components found during the work.

Component	Issue	Finding
ROS 2 rosidl	`file(STRINGS)` without `ENCODING UTF-8` breaks non-ASCII workspace paths	F11
CUDA 13.2 CCCL	specialization outside its namespace, rejected by GCC 13/14	F22
ONNX Runtime	released aarch64 CUDA EP has no `sm_87` code; release CMake configuration paths	F8, F9
SAM	`CMAKE_CUDA_ARCHITECTURES` set unconditionally without 87; `ekf_core` not installed under Lyrical; hard-coded plugin path	F18, F55
Lyrical tf2	`tf2/LinearMath/Quaternion.h` removed (only `.hpp`)	F28
ONNX Runtime C++ API	`GetInputTypeInfo(i).GetTensorTypeAndShapeInfo()` is a view into a temporary	F41

11 Implementation process

11.1 Timeline

Figure 16 shows the phases. **Phase 0** (decisions): a first scaffold from 2026-10-26 had never been built or tested and was moved to `attic/`. The decisions recorded in `PLAN.md` fixed the boundary conditions: boards stay on stock JetPack (D1), Lyrical runs both in a container and natively from NVIDIA's Buildfarm (D2), and C++ is used for the later agent and GUI (D5).

Phase 1 (build side, 2026-11-02). The order of work followed the dependency chain: image, sysroot builder, toolchain, test suite, then the real workload. Each layer was accepted only when its probe test passed (CUDA, then TensorRT, ONNX Runtime, ROS interfaces, a CUDA ROS node). The SAM workspace exposed the remaining problems (Kilted APIs, acados code generation, the UTF-8 path bug), and ONNX Runtime had to be built from source for `sm_87`.

Phase 1b (board side, 2026-11-03). With SSH access to the Nano, the cross-built tests ran 8/8 after one fix to the board runner (sourcing ROS setup scripts without `set -u`, `F36`, `F37`). The board-side features followed from concrete needs: shipping a ROS distribution (to run Kilted on a Jazzy board), the library closure check (after an Isaac test binary failed), debugging, tracing, the launch mode, container mode and finally the measurement campaigns of Section 9, which in turn found the undeclared dependencies and lifecycle problems of Section 10.

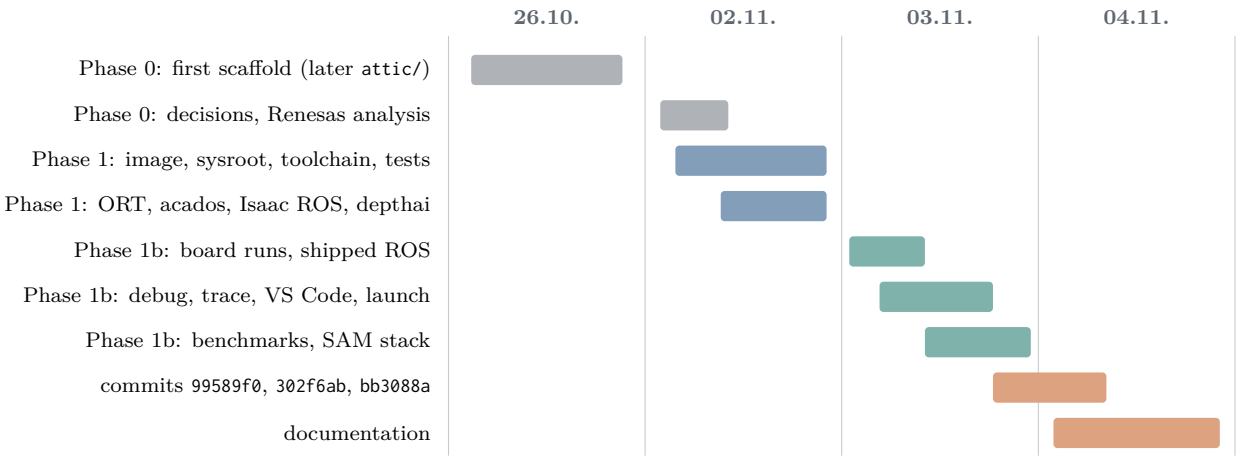


Figure 16: Implementation timeline (days with activity; the columns are not to scale).

11.2 Method

- **Evidence-driven engineering.** Every non-obvious observation was recorded as a numbered finding together with its consequence for the system before the next step; the findings log serves as design rationale and as regression list.
- **Probe-first verification.** Each capability has a minimal probe that exercises exactly one layer, so a failure points to its cause; only then is the full SAM workspace built.
- **Static checks of generated shell.** `jxb` composes nested shell scripts (lock wrapper, container wrapper, environment, launch script). During development they were checked with `bash -n` through a dry-run harness that replaced the command runner and recursively checked every nested `bash -c` script.
- **Real hardware as the final test.** QEMU runs catch instruction-set and linkage errors early, but every feature was accepted only after it ran on the Nano.
- **Agent-assisted, parallel development.** The work was carried out with AI coding agents (Claude Code) under the direction of the project owner: a main development session, forked

sessions for reviews and long measurement campaigns, and a separate hardware-in-the-loop session that shared the board. Coordination used explicit ownership of files and measurements and, on the board, the measurement lock, which the HIL session adopted. A forked review found the process-group issues of Section 6.6 (F58).

11.3 Incidents

- **Out-of-memory builds** (exit 137) on the 15 GB host, resolved by bounding the parallelism (F54).
- **The board reset** of F56, which also interrupted other sessions on the shared board.
- **Lost work** when a session restored an earlier checkpoint and overwrote files; the changes were re-applied from the session transcript. Lesson: commit early on shared working trees.

12 Limitations and future work

Open issues of the current state:

- **Kernel tracing on the board** has been tested only against a mock tracefs; a real run needs root, i.e. a sudo password typed by the user or explicit approval of the privileged-container route on the shared board.
- **The reset of F56** is unexplained (pstore needs root).
- **Lyrical runtimes** (fixed 2026-11-04): `ros2 bag record` needs `--topics` there (F68), and SIGINT has to be restored before `ros2 launch`, because Ubuntu 26.04's `timeout` passes it on ignored (F69).
- **Single board class.** Only an Orin Nano was available; AGX Orin and Thor (GPU variant `openrm`) are prepared in the profile format but untested.
- **Missing pieces of the SAM stack**: the Basler Pylon SDK, ros2_control and Gazebo (a size decision for the sysroots), and CUDA kernel debugging with `cuda-gdb`.

Planned next steps (`PLAN.md`): an `orin-agent` daemon with a stable API (health, deploy, run, logs, trace) as the common back end of jxb and the later JetsonPlanner GUI; multi-board deployment and synchronized traces on several boards with Perfetto as the common trace format; a provisioning bundle (apt repository with cross-built packages, container images, configuration) on top of stock JetPack, versioned together with the sysroot's package lock; and CI for images, sysroots and the probe tests.

13 Conclusion

jxb shows that a correct cross-development environment for Jetson Orin can be *constructed* rather than copied. A private apt instance resolves the board's package closure from the board's own repositories; extraction plus a small, explicit emulation of maintainer-script effects yields a sysroot that matches the board by construction; and a toolchain whose defaults are pinned to the board compiles C++, CUDA and Python extensions without emulation. The same sysroot doubles as the root file system of a GPU-enabled runtime container, which lets one stock board run three ROS distributions natively or in a container. On top, a launch mode with process-group lifecycle management and an observability pipeline with machine-readable summaries turn the board into a measurable system. Most of the effort went into details that are invisible when building natively, such as symbolic-link semantics, `DT_NEEDED` records, generated paths, undeclared dependencies and process lifecycles; documenting them, as this report and the findings log do, is what makes the system maintainable.

A Command reference

Command	Purpose
`jxb image build | runtime`	build the xbuild image / the board runtime image of the profile
`jxb sysroot build [--add PKG ...] [--no-update]`	build the sysroot from the board's repositories
`jxb sysroot rosdeps -w WS`	workspace rosdeps and launch-file dependencies as package set
`jxb sysroot info`	metadata of the current sysroot
`jxb up [-w WS ...] [--recreate]`, `jxb down`	create / remove the dev container
`jxb shell`, `jxb exec -- CMD`	shell or command in the dev container
`jxb build [colcon args]`	`cross-colcon-build` in the current workspace
`jxb test [--board B]`	build and check the test suite, optionally run it on a board
`jxb board list | check B`	inventory; compare a board's package versions with `packages.lock`
`jxb deploy B [--ws WS] [--packages P ...] [--no-files]`	deploy a workspace
`jxb run B -- CMD`, `jxb ssh B`	command or shell on the board with the deployment's environment
`jxb gui B -- PROGRAM`	a board program as a window here (X11 over SSH)
`jxb launch B [--packages ...] [--monitor] [--bag [T ...]] [--with CMD] [--duration S] -- TARGET`	deploy and launch, optionally recorded
`jxb debug B --pkg P --exe E [--launch PKG FILE] [--server-only] [--cuda]`	remote debugging (`--cuda`: cuda-gdbserver and cuda-gdb)
`jxb gdb [--cuda] [-- ARGS]`	cross gdb (or cuda-gdb) for the profile
`jxb trace nsys | tegrastats | kernel | bench B ...`	recordings on the board
`jxb trace report | plot | compare | perfetto | list | gui`	analysis on the host; `perfetto`: one timeline of recordings
`jxb vscode init -w WS [--board B ...] [--pkg P --exe E]`	VS Code workspace (several `--board`: debugging on all of them at once)

Global options: `--profile NAME`, `--mode auto|native|container`, `--dry-run` (prints the commands instead of running them). The board argument of `deploy`, `run`, `launch`, `test --board`, `board check` and the recordings also takes `B,B`, `@GROUP` or `all`: the command runs on those boards at once, with a result table.

B Configuration reference

Key	Meaning
Profile (`profiles/<name>.toml`)	

Key	Meaning
inherit	parent profile; tables merge recursively
[board] jetpack, l4t, soc, cpu_flags, cuda_arch, gpu_variant, ros_distro, ros_mode, qemu_cpu	the board's software and hardware
[image] tag, ubuntu, ros_distro, ros_apt_url, cuda_version, cuda_pkg_version	build arguments of the xbuild image
[sysroot] dir, codename, python, cuda_target, sources, preferences, package_sets, extras, rosdep_fallback_codename	the sysroot recipe
[keys]	apt keyrings (paths or URLs)
[extras.<name>]	tarball, debs or url; dest, optional, build
[container] board_paths	board paths mapped into the dev container
[deploy] runtime, container	shipped runtime extras; forced container mode
Boards (boards.toml, local)	
[defaults], [boards.<name>]	host, user, ssh_port, deploy_dir, codename, gdb_port, nsys, trace_dir, measure_lock, trt_cache, cuda_gdbserver, groups (for @GROUP), ros_domain_id
Local defaults (jxb.toml, local)	
profile, mode, workspaces	defaults for --profile, --mode and the dev container
[sysroot] extra_sets	additional package sets, e.g. ws-{profile}-<ws>
[workspace.<dir>] files, env, container_env	deployed source-tree files; board environment; extra variables in container mode

C The summary.json schema

Key	Present for	Content
trace, dir, meta	all	name, directory, meta.json
board_state	all (since 2026-11-04)	start, end: nvpmodel; cpu.<policy> (governor, min_mhz, max_mhz, cur_mhz, hw_min_mhz, hw_max_mhz); devfreq.<device> (governor, min_mhz, max_mhz, cur_mhz; GPU, EMC = bwmgr, engines); cpuidle (states, disabled); gpu_railgate; fan; temp_c; fixed (clocks with minimum = maximum, disabled idle states); changed (settings that differ between start and end)
nsys	nsys	span_s; gpu (kernel_launches, kernel_time_ms, busy_ms, busy_pct, top_kernels); memcpy; cuda_api; nvtx; os_runtime; cpu_samples; processes
tegrastats	monitored runs	samples, interval_ms, duration_s, metrics.<key>.{mean,min,max}, energy_j, csv
processes	--monitor	session_cpu_pct, session_rss_mb, monitor_cpu_pct; session[], top[] with cpu_share_pct, cpu_mean_pct, cpu_max_pct, rss_max_mb, series

Key	Present for	Content
`launch`	launch	`processes`, `finished_cleanly`, `died`, `crashed`, `errors`, `warnings`, `first_errors`
`bag`	`--bag`	topics with type, message count and rate; CSV series of numeric topics
`kernel`	kernel	`span_s`, `events`, `ctxsw_per_s`, `cpu_busy_pct`, `top_tasks` (with `wakeup_us`), `top_processes`, `irqs`
`bench`	bench	per model: runs sorted by p50 with `p50_ms`, `p99_ms`, `fps`, `speedup_vs_ort_cuda`

D FAQ and known issues

Why not just build on the board? It works for small packages, but the Orin Nano has 8 GB of memory shared between CPU and GPU and six cores; building there also requires `-dev` packages on a board that should stay stock.

Why not QEMU? Emulated compilation is roughly an order of magnitude slower, and `nvcc` under emulation was unreliable (P2). `jxb` uses QEMU only for optional smoke tests.

Can I use my own CMake project without colcon? Yes: in the dev container, run `source /opt/jxb/toolchain/env.sh` and configure with `-DCMAKE_TOOLCHAIN_FILE=$JXB_TOOLCHAIN_FILE`.

`jxb board check` shows `DIFF`. Patch-level differences within a release are ABI-compatible (`F35`). Major differences mean that the board is not on the profile's JetPack release.

A node finds the wrong ONNX Runtime. `jxb`'s GPU build comes first on `LD_LIBRARY_PATH` only in `jxb`'s environment (`jxb run`, `jxb launch`, `jxb ssh`); a plain SSH shell sees the board's CPU-only `/opt/onnxruntime`.

Known issues. ONNX Runtime's CUDA EP in container mode (`F56`); `pylon_camera_package` needs the Basler SDK; `micro_ros_agent` fails on Kilted because its vendored Fast DDS references `tinyxml2::tinyxml2` (`F54`).

E Glossary

Term	Meaning
ABI	application binary interface: calling convention, data layout and symbol versions that binaries rely on
CDI	Container Device Interface, a specification of device nodes and mounts for containers
closure	the set of packages reachable through dependencies from the requested ones
EP	ONNX Runtime execution provider (CPU, CUDA, TensorRT)
fp16, INT8, Q/DQ	half precision; 8-bit integer; explicit quantize/dequantize nodes in an ONNX model
L4T	Linux for Tegra, NVIDIA's board support package (Jetson Linux)

Term	Meaning
merged `/usr`	layout in which `/bin`, `/lib` and `/sbin` are symbolic links into `/usr`
PREEMPT_RT	the real-time preemption patch set of the Linux kernel
SBSA	Arm Server Base System Architecture; NVIDIA's unified Arm CUDA target layout
`SONAME`	the name a shared library declares for itself, recorded by the linker in `DT_NEEDED`
sysroot	a directory that compiler and linker treat as the target's root file system
underlay, overlay	an installed ROS environment; a workspace built on top of it

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